PLAIN-ENGLISH COMPANION TO THE OUTBOUND NETWORK DESIGN PACKET

Order to door, in plain English

One package, happy path. One whole day, happy path. Then the things that go wrong, one at a time. Then the measurements, the weekly scorecard, how this job would build the dashboard, and transportation planning: what is in a trailer, what is in a mile, and how to buy service with trucks.

Who this is for. Rajesh Rishiyur Chandrasekaran, preparing for a role that owns the design of a large outbound transportation and sorting network. Written so that someone with no logistics background could follow every step and every calculation.

How it is written. No abbreviations. Every calculation is shown as arithmetic you can check with a pencil. We use one made-up network with one set of numbers, stated in full in Chapter 1, and we never change them. We walk one customer's package first with nothing going wrong. Then one whole day with nothing going wrong. Only after that do we add problems, one at a time, and show what each one costs and how you would measure it. The measurements are explained from that same example, so there is nothing to memorize.

What is real and what is made up. The shape of the journey (warehouse, truck, sorting hub, truck, neighborhood depot, van) is how Amazon has publicly described its network; the sources are in Chapter 9. Every rate, time, and dollar amount is made up for the example so the arithmetic is clear. None of them is an Amazon figure. Chapter 9 also gives the words you will hear in the interview for the plain words used here.

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How to read this. Read Chapters 1 to 3 in order with a pencil; recompute three or four of the boxes yourself. Chapter 4 can be read one problem at a time. Chapter 5 should make sense without effort once Chapter 4 is done, because every measurement is a problem from Chapter 4 turned into a fraction. Chapters 6 and 7 are what the job produces every week. Chapter 8 goes one level down into the trucks, in the same question-arithmetic-answer style: what fills a trailer, how full to plan it, what is in a mile, when a lane deserves its own truck, how many tractors and drivers, how to buy service, the transportation scorecard, the optimization model in the companion workbook, and a camera that measures the air inside a gaylord. The long, detailed packet in the same folder covers the same ground with more depth; this one is the explanation to read first.

Chapter 1 — Our pretend network and every number we will use

Here is the whole network. A package goes through six places in order. We give each a plain name and use only that name.

```
THE WAREHOUSE  ──truck, 200 miles──▶  THE SORTING HUB  ──truck, 120 miles──▶  DEPOT 7  ──van──▶  MARIA'S DOOR
(items are picked,                    (packages from many                    (packages are sorted
 boxed, labeled,                        warehouses are sorted                   by street and loaded
 and loaded)                            by destination depot)                   into vans)
```

Below is every number we will use, in one place. Each one is made up, but each is the kind of number a real operation would know. If you want to change one, change it here and the rest of the document changes with it.

THING	NUMBER	WHAT IT MEANS, IN ONE SENTENCE
People and pay		
Hourly pay, every building	$22 an hour	Wages plus benefits for a warehouse, hub, or depot worker. One number everywhere, to keep the arithmetic simple.
Supervisors, breaks, and helpers	15% on top	For every 100 hours of hands-on work, the building pays about 115 hours, because people take breaks and someone supervises and fixes problems. We multiply hands-on hours by 1.15.
Shifts	Two shifts of 10 hours	The warehouse runs 20 hours a day.
The warehouse		
Packages shipped a day	100,000	What the warehouse sends out on a normal day.
Items per package	1.6 on average	Some orders are one item, some are three. Maria's order is two items.
Picking speed	160 items an hour	One worker finds and picks 160 items an hour from the shelves (in a modern warehouse the shelves come to the worker).
Packing speed	100 packages an hour	One worker boxes and tapes 100 packages an hour.
Box, tape, and padding	$0.15 a package	Materials.
Labeling machines	3 machines, 2,500 an hour each	Each package is weighed, checked, and gets its shipping label here. Together: 7,500 an hour.
Sorting machine (warehouse)	10,000 an hour	Reads the label and sends the package down a chute toward the right truck door.
Loading speed	500 packages an hour	One worker fills big cardboard bins and rolls them into the trailer.
Truck leave times	4 a day: 3 AM, 9 AM, 3 PM, 9 PM	Four times a day a set of trailers leaves for the hub. About 25,000 packages ride each leave time.
Time from last pack to truck leaving	2 hours	A package must be packed two hours before its truck leaves, to be labeled, sorted, and loaded.
Time from order to packed	Usually 1.2 hours; 9 orders in 10 within 2 hours	How long the warehouse takes to turn an order into a labeled package.
Trucks		
Trailer, full	2,000 packages	What fits if every spot is used.
Trailer, planned load	1,600 packages (80% full)	We plan to fill 80% and leave room, because the number of packages each day varies. Chapter 4.2 explains why 80 and not 100.
Truck cost	$3.20 a mile	Driver, fuel, tractor, trailer, insurance, and the empty trip back, all in.
Warehouse to hub	200 miles; 4 hours usually	200 × $3.20 = $640 a trip. Some trips take longer; Chapter 4.1 deals with that.
Hub to Depot 7	120 miles; 2.5 hours	120 × $3.20 = $384 a trip.
The sorting hub		
Packages a day	100,000	From our warehouse and others; four sorting windows of 3 hours, about 25,000 packages each.
Sorting windows	1–4 AM, 7–10 AM, 1–4 PM, 7–10 PM	Trucks arrive at the start of a window; depot trucks leave at the end of it.
Truck doors	12	Twelve trailers can be unloaded at once.
Unloading speed	400 packages an hour per worker; 2 workers per door	A 1,600-package trailer takes 1,600 ÷ 800 = 2 hours to unload.
Scanning speed	1,000 an hour per station	Each package is scanned onto the sorting machine.
Sorting machine (hub)	20,000 an hour	Sends each package to the chute for its depot.
Bagging and loading speed	600 an hour per worker, each	Packages for one depot are put into bags; bags are loaded onto that depot's trailer.
Hub fixed costs	$13,000 a day	Rent, the sorting machine's depreciation, electricity, salaried managers. Paid whether 60,000 or 100,000 packages come through.
Wrong-building rate	3 in 1,000	Packages the hub sends to the wrong depot. Each costs about $2 to put right.
Depot 7 (one of about 17 depots the hub serves)		
Packages a day	6,000	Arrives overnight on about four trailers.
Last trailer must arrive by	6:30 AM	Anything later misses the morning vans.

Unload, sort, stage speeds	600, 250, 400 an hour per worker	Sorting here means putting each package in the right van's bag, in the order of the stops.
Depot fixed costs	$560 a day	Rent, lights, a manager.
Vans leave	9 AM	Thirty vans, one wave. There is also a smaller afternoon wave of gig drivers for late packages, which costs more (next row).
Vans		
Packages per van	200	One van, one driver, one day.
Stops per van	160	200 packages ÷ 1.25 packages per address = 160 doors.
Van day	8 hours driving; deliveries between 9 AM and 8 PM	160 stops ÷ 8 hours = 20 stops an hour, one every three minutes.
Van cost	$33 an hour → $264 a day	$22 driver + $8 van + $3 fuel and other, × 8 hours.
Afternoon gig drivers	$1.90 a package	Drivers in their own cars, paid by the block, for packages that missed the morning vans. Costs more than $264 ÷ 200 = $1.32 in a regular van.
Outside carrier (postal service or similar)	$4.50 a package	For packages going somewhere our vans do not go; we hand them over at the warehouse dock.

Two words we will use constantly. A *pile* is packages waiting somewhere: in a bin at the door, in a trailer in the yard, on the hub floor. A *deadline* is a clock time by which the pile must be empty: the truck leaves at 9 PM; the hub window closes at 4 AM; the vans leave at 9 AM. Almost everything that goes wrong in this business is a pile that was not empty when its deadline came. Almost everything that costs too much is a truck, a building, or a van that carried fewer packages than it was paid for. Keep those two sentences in mind and the rest follows.

Chapter 2 — One package, nothing goes wrong: Maria's kettle, order to door

Maria lives in Depot 7's neighborhood. On Monday at 2:12 PM she orders a kettle and a box of tea. The website says "arrives Tuesday." Here is everything that happens to her package, with the clock, and with the cost of each step worked out on the spot.

2.1 The journey, step by step, with the clock and the cost

How we work out the cost of a step: take what the worker (or truck) costs per hour, divide by how many packages that worker (or truck) handles in that hour. That gives the cost of one package's share. Every row uses only that idea.

CLOCK	WHAT HAPPENS	THE ARITHMETIC FOR THIS STEP	THIS STEP COSTS	COST SO FAR
Mon 2:12 PM	Maria clicks "buy." The computer picks our warehouse (it has both items and is the closest), and assigns the order to the **9 PM truck** to the hub. The website shows "arrives Tuesday" because the clock-backward calculation in 2.2 says it can.	No labor yet. Software.	$0.00	$0.00
2:20 PM	The order is released to a picker along with 11 other orders, so one walk collects 20 items.	Waiting in the computer: 8 minutes.	$0.00	$0.00
2:48 PM	The kettle and the tea are picked into a plastic tote.	Picker costs $22 an hour and picks 160 items an hour. One item = $22 ÷ 160 = $0.1375. Maria has 2 items: 2 × $0.1375 = $0.275.	$0.28	$0.28
3:10 PM	The tote rides a conveyor to a packing station. The two items are put in one box with padding and taped.	Packer costs $22 an hour and packs 100 packages an hour: $22 ÷ 100 = $0.22. Box, tape, padding: $0.15.	$0.37	$0.65
3:14 PM	**Labeling.** The box is weighed (1.9 pounds, matching what a kettle and a tea box should weigh, so nothing is missing), the shipping label prints and is stuck on, and the package is recorded as "exists, going to Depot 7, on the 9 PM truck." This is the moment the package becomes a package.	Machine time is tiny. The people who run the labelers, the sorter, and fix jams come to about $0.09 per package (Chapter 3.1 shows the hours). Supervisors and breaks add 15% to the labor above: 0.15 × ($0.275 + $0.22 + $0.09) ≈ $0.09.	$0.18	$0.83 → call it **$0.80**
3:18 PM	The warehouse sorting machine reads the label and drops the package down the chute for "hub, 9 PM." It lands in a big cardboard bin at door 14.	Included in the $0.09 above.	—	$0.80
7:40 PM	The bin is full, closed, and rolled into trailer 14-B. Maria's package has waited at the door 4 hours 22 minutes. That is the longest it will wait anywhere in a building, and it is on purpose: the truck leaves at a fixed time so it leaves full.	Loader costs $22 an hour and loads 500 an hour: $22 ÷ 500 = $0.044. Already inside the $0.09 above.	—	$0.80
9:00 PM	**The truck leaves.** Doors close at 9:00 with 1,590 packages aboard (plan was 1,600). Sealed, pulls out at 9:05.	Trip costs 200 miles × $3.20 = $640. Share per package: $640 ÷ 1,590 = $0.4025.	$0.40	$1.20
Tue 1:17 AM	Arrives at the hub after 4 hours 12 minutes on the road, a normal trip. A door is free. Unloading starts at 1:20.	Two unloaders at 400 an hour each = 800 an hour. 1,590 packages ÷ 800 = 2 hours to empty the trailer.	—	$1.20
2:05 AM	Maria's bin comes off. 2:10 scanned onto the hub's sorting machine. 2:11 the machine drops it in the **Depot 7** chute. 2:40 it is put in a Depot 7 bag. 3:30 the bag is loaded onto the Depot 7 trailer.	Hub cost per package is $0.30: $0.17 of labor (unload, scan, bag, load; Chapter 3.3 shows the hours) and $0.13 of the hub's fixed costs ($13,000 a day ÷ 100,000 packages).	$0.30	$1.50
4:00 AM	**Window closes; the Depot 7 trailer leaves** with 1,600 packages.	120 miles × $3.20 = $384. $384 ÷ 1,600 = $0.24.	$0.24	$1.74
6:25 AM	Arrives at Depot 7, five minutes before the 6:30 deadline. Unloaded by 6:50. Scanned 7:05. The computer puts it on **van route 17, stop 92 of 160**. By 8:30 it is in route 17's bag, in stop order.	Depot cost per package is $0.30: $0.21 labor (Chapter 3.4) and $0.09 of fixed cost ($560 ÷ 6,000).	$0.30	$2.04
9:04 AM	**Van 17 leaves** with 198 packages for 158 stops.	Van day costs $264. $264 ÷ 198 = $1.333.	$1.33	$3.37
1:53 PM	**Delivered.** Stop 92 is reached after 91 earlier stops at 20 an hour (4 hours 33 minutes) plus a 15-minute break. Photo of the package on the porch. Promise kept: "arrives Tuesday."	—	—	**$3.37**

Check the total. Warehouse $0.80 + first truck $0.40 + hub $0.30 + second truck $0.24 + depot $0.30 + van $1.33 = $3.37. The average package costs a little less ($3.33), because Maria's order had two items and the average has 1.6. From the warehouse dock to Maria's door, leaving out the warehouse's own work, the cost was $3.37 − $0.80 = **$2.57**. That dock-to-door number is the one this job is measured on. It has a formal name, *transportation cost per package*, and Chapter 5.1 defines it word by word.

2.2 Why the website said "arrives Tuesday": the clock runs backward

Nobody guessed Tuesday. The computer started from the end and subtracted. Each line below is "the step before must finish by this time, because the next step needs this long."

```
THE CLOCK, RUN BACKWARD FROM MARIA'S DOOR

 Tue  8:00 PM   Latest the van can still be making deliveries ("arrives Tuesday" means by 8 PM).
 Tue  9:00 AM   Vans leave Depot 7.                          (A van needs its full day: 160 stops at 20 an hour = 8 hours, plus breaks.)
 Tue  6:30 AM   Last trailer must ARRIVE at Depot 7.         (Depot needs 2.5 hours: unload 0.5, sort 1.3, put in van bags 0.5, spare 0.2.)
 Tue  4:00 AM   Trailer must LEAVE the hub.                  (The drive is 2.5 hours.)
 Tue  1:00 AM   Our truck must ARRIVE at the hub.            (Hub needs 3 hours: unload 2, sort and bag 0.4, spare 0.6.)
 Mon  9:00 PM   Our truck must LEAVE the warehouse.          (The drive is usually 4 hours.)
 Mon  7:00 PM   Maria's package must be PACKED.              (2 hours to label, sort, and load.)
 Mon  5:00 PM   Maria must ORDER by now.                     (9 orders in 10 are packed within 2 hours of ordering.)

 Maria ordered at 2:12 PM, nearly three hours before the 5 PM deadline, so the website showed "arrives Tuesday."
 Had she ordered at 5:30 PM, her package would have been assigned to the 3 AM truck, and the website would have said Wednesday.
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Three things to notice. First, every deadline on this list is set by the deadline after it; move the van departure to 8 AM and every line above it moves an hour earlier, including the time Maria must order. Second, the 9 PM truck departure is the most important deadline in the warehouse: it is what every pile inside the building is racing. (The industry word for it is *critical pull time*. This document just says "the truck leaves at 9 PM.") Third, the "spare" minutes in the brackets add up to about 5 hours across the whole chain. That spare time is what absorbs a slow trip or a slow unload. Chapter 4 is mostly about what happens when it runs out.

2.3 Where the twenty-four hours went, and where the $3.37 went

The time

WHAT THE PACKAGE WAS DOING	ADDING IT UP	HOURS	SHARE
Being worked on by a person or machine	Pick ~2 min + pack 3 + label 0.2 + sort 0.1 + unload share 2 + hub scan and sort 1 + bag 1 + depot sort 1 + load van 2 + the delivery stop 1	about 13 minutes	1%
Riding in a truck or van	4 h 12 min + 2 h 25 min + 4 h 49 min	11 h 26 min	48%
Waiting for a scheduled departure (the bin at door 14, the hub trailer, the van bag)	5 h 47 min + 30 min + 34 min	6 h 51 min	29%
Waiting in a line (for a picker, a packer, a door, a scanner)	Everything else	about 5 h 10 min	22%
Order to door	**Mon 2:12 PM to Tue 1:53 PM**	**23 h 41 min**	**100%**

The package was touched for thirteen minutes in a day. Almost half the time it was on a road. Nearly a third of the time it was waiting for a scheduled truck or van to leave. That waiting is not waste: it is what makes the truck full, and a full truck is what makes the trip cheap. The part a designer attacks is the fourth line, waiting in a line, because that is where the spare time disappears on a bad day.

The money

WHO WAS PAID	ARITHMETIC	DOLLARS	SHARE OF DOCK-TO-DOOR ($2.57)
The warehouse (pick, pack, box, label, load)	Chapter 2.1 rows 3–5	$0.80	not included
First truck	$640 ÷ 1,590	$0.40	16%
The hub	$0.17 labor + $0.13 fixed	$0.30	12%
Second truck	$384 ÷ 1,600	$0.24	9%
The depot	$0.21 labor + $0.09 fixed	$0.30	12%
The van	$264 ÷ 198	$1.33	52%
Total		**$3.37**	**dock-to-door $2.57**

Half of the dock-to-door money is the van. The two trucks together are a quarter. The two sorting buildings together are a quarter. That is why, later, "how many packages ride in each van" will turn out to be the single most expensive number in the network.

IF SOMEONE ASKS YOU TO EXPLAIN THIS IN ONE BREATH — 109 WORDS

Maria orders at two in the afternoon. By three her two items are picked, boxed, and labeled, and the label commits the package to the nine o'clock truck. It waits at the door until the truck leaves, rides four hours to a sorting hub, is sorted into a bag for her neighborhood depot, rides two and a half hours there, is sorted again into one van's bag in stop order, and the van leaves at nine. She has it by two the next afternoon. Thirteen minutes of work, eleven hours of road, seven hours waiting for departures. It cost three dollars thirty-seven, and half of that was the van.

Chapter 3 — One whole day, nothing goes wrong: 100,000 packages

Same network, same numbers, but now the whole day. The question at every step is the same three-part question: how many hours of work, how many people or trucks or doors does that need, and what does it cost per package. Each box shows the arithmetic in full.

3.1 Inside the warehouse: people, hours, and dollars

Picking

STEP	QUESTION	ARITHMETIC	ANSWER
1	How many items must be picked?	100,000 packages × 1.6 items each	160,000 items
2	How many hours of picking?	160,000 ÷ 160 an hour	1,000 hours
3	How many hours do we pay for, with breaks and supervisors?	1,000 × 1.15	1,150 hours
4	How many pickers on each 10-hour shift?	1,150 ÷ 2 shifts ÷ 10 hours	57.5, so 58 people
5	What does picking cost today?	1,150 hours × $22	$25,300
6	Per package?	$25,300 ÷ 100,000	$0.25

Packing

STEP	QUESTION	ARITHMETIC	ANSWER
1	Hours of packing?	100,000 ÷ 100 an hour	1,000 hours
2	Paid hours?	1,000 × 1.15	1,150 hours; 58 packers a shift
3	Cost?	1,150 × $22	$25,300 → $0.25 a package
4	Boxes, tape, padding?	100,000 × $0.15	$15,000

Labeling, sorting, loading

STEP	QUESTION	ARITHMETIC	ANSWER
1	Can the labeling machines keep up?	Average: 100,000 ÷ 20 hours = 5,000 an hour. Busiest hour is about 1.4 times average = 7,000. Machines do 3 × 2,500 = 7,500.	Yes, but at the busiest hour they are 7,000 ÷ 7,500 = 93% busy. That is close to the edge; Chapter 4.5 shows what happens when it tips.
2	Can the sorting machine keep up?	7,000 at the busiest hour ÷ 10,000	70% busy. Comfortable.
3	Hours of loading?	100,000 ÷ 500 an hour = 200; × 1.15	230 paid hours
4	People running the labelers and the sorter and fixing problems?	3 labeler operators × 20 hours + 2 sorter operators × 20 hours + 100 hours of problem-solving	200 paid hours
5	Cost of this group?	(230 + 200) × $22	$9,460 → $0.09 a package

The warehouse total

ITEM	ARITHMETIC	ANSWER
Picking + packing + labeling/sorting/loading + boxes	$25,300 + $25,300 + $9,460 + $15,000	$75,060 a day
Per package	**$75,060 ÷ 100,000**	**$0.75**

3.2 The trucks to the hub

How many trailers, and what they cost

STEP	QUESTION	ARITHMETIC	ANSWER

1	Packages per truck leave time?	100,000 ÷ 4 leave times	25,000
2	Trailers per leave time at the planned 1,600 each?	25,000 ÷ 1,600	15.6, so 16 trailers
3	Trailers a day?	16 × 4	64
4	Cost a day?	64 × $640	$40,960
5	Per package?	$40,960 ÷ 100,000	$0.41
6	Why $0.41 and not the $0.40 we used for Maria?	64 trailers carried 100,000, so the average trailer had 100,000 ÷ 64 = 1,563 packages, a little under the 1,600 plan. $640 ÷ 1,563 = $0.41.	One cent, because the trailers were 2% emptier than planned. Keep this in mind for Chapter 4.2.
7	How many tractors (the truck that pulls the trailer)?	A round trip is 4 hours out + 2 hours being unloaded + 4 back + 1 in the yard = 11 hours, so one tractor does about 2 trips a day. 64 ÷ 2 = 32, plus 10% for repairs.	About 35 tractors
8	How many trailers (the box) in the pool?	In the cycle: 64 × 11 ÷ 24 = 29. Plus 16 being loaded at the warehouse and 16 waiting at the hub.	About 60 trailers. This is why changing a truck schedule takes weeks, not minutes: the boxes have to be somewhere.

3.3 The sorting hub

The hub takes 100,000 packages a day in four 3-hour windows. In each window about 16 trailers arrive, are unloaded, every package is scanned and sent down the chute for its depot, the chutes are bagged, and the bags go onto about 16 outgoing trailers, one or more per depot. When the window closes at the end of the 3 hours, the outgoing trailers leave. The window is the deadline.

Doors, people, and the machine, for one window of 25,000 packages

STEP	QUESTION	ARITHMETIC	ANSWER
1	How long to unload one trailer?	1,600 ÷ (2 people × 400 an hour)	2 hours
2	How many door-hours does the window need?	16 trailers × 2 hours	32 door-hours in a 3-hour window
3	How many doors?	32 ÷ 3 = 10.7 if the trailers arrived perfectly spaced. They do not. Allow for bunching.	12 doors. (Chapter 4.1 shows a night when 12 is not enough.)
4	How many scanning stations?	25,000 ÷ 3 hours = 8,333 an hour; ÷ 1,000 per station = 8.3; people are not 100% busy, so ÷ 0.85	10 stations
5	Is the sorting machine big enough?	8,333 an hour ÷ 20,000	42% busy. It is sized for the busiest season, not a normal day.
6	How many people bagging, and how many loading?	8,333 ÷ 600 an hour	14 bagging and 14 loading during the window

The hub's cost for the day

STEP	QUESTION	ARITHMETIC	ANSWER
1	Hands-on hours?	Unload 100,000 ÷ 400 = 250; scan 100,000 ÷ 1,000 = 100; bag 100,000 ÷ 600 = 167; load 100,000 ÷ 600 = 167	683 hours
2	Paid hours and cost?	683 × 1.15 = 786 hours × $22	$17,288 → $0.17 a package
3	Fixed costs per package?	$13,000 ÷ 100,000	$0.13
4	**Hub cost per package**	**$0.17 + $0.13**	**$0.30 ($30,288 a day)**
5	What if only 60,000 packages came through?	Labor stays $0.17 a package (fewer hours); fixed becomes $13,000 ÷ 60,000 = $0.22	$0.39 a package. A sorting building gets expensive when it is under-used, because the rent does not shrink.

3.4 The trucks to the depots, and Depot 7

Second truck

STEP	QUESTION	ARITHMETIC	ANSWER
1	Cost of one trip hub → Depot 7?	120 miles × $3.20	$384

2	Per package at the planned 1,600?	$384 ÷ 1,600	$0.24
3	Trailers a day from the hub to all depots?	100,000 ÷ 1,600 = 62.5; in practice 64	64 trailers; 64 × $384 = $24,576 a day
4	How many does Depot 7 get?	6,000 ÷ 1,600 = 3.75	4 trailers a night; the fourth is lighter

Depot 7: 6,000 packages, 30 vans

STEP	QUESTION	ARITHMETIC	ANSWER
1	How many sorters in the 4 AM–9 AM window?	6,000 ÷ 5 hours = 1,200 an hour; ÷ 250 per sorter	4.8, so 5 people
2	Does the last trailer (arriving 6:30) make the 9 AM vans?	Unload 30 min; sort 1,600 ÷ (5 × 250) = 1.28 hours = 77 min; into van bags 30 min. 6:30 + 0:30 + 1:17 + 0:30	Ready at 8:47. Thirteen minutes to spare. The 6:30 deadline is real.
3	Hands-on hours?	Unload 6,000 ÷ 600 = 10; sort 6,000 ÷ 250 = 24; bag and load vans 6,000 ÷ 400 = 15	49 hours
4	Cost?	49 × 1.15 = 56.4 hours × $22 = $1,240; ÷ 6,000	$0.21 a package labor
5	Fixed?	$560 ÷ 6,000	$0.09
6	**Depot cost per package**	**$0.21 + $0.09**	**$0.30. Assume every depot is like Depot 7: $0.30 × 100,000 = $30,000 a day.**
7	Vans?	6,000 ÷ 200	30 vans, 30 drivers, at 9 AM

3.5 The vans

What a van costs and what a package pays

STEP	QUESTION	ARITHMETIC	ANSWER
1	A van for a day?	8 hours × ($22 driver + $8 van + $3 fuel and other)	$264
2	Stops per van, and per hour?	200 packages ÷ 1.25 per address = 160 stops; 160 ÷ 8 hours	20 stops an hour, one every 3 minutes
3	Cost per package?	$264 ÷ 200	$1.32
4	If the van carried 190? 180? 220?	$264 ÷ 190; ÷ 180; ÷ 220	$1.39; $1.47; $1.20. Ten packages either way is about seven cents on every package.
5	Vans across the whole network?	100,000 ÷ 200 = 500 vans; × $264	$132,000 a day
6	Could a van take 260 packages?	260 ÷ 1.25 = 208 stops ÷ 20 an hour = 10.4 hours of stops	No: the delivery window is 9 AM to 8 PM, 11 hours, and the driver needs breaks and the drive back. 220 (8.8 hours) is about the limit.

3.6 The day's bill, and the two numbers everyone asks for

Adding up the day

WHO WAS PAID	FROM	DOLLARS	PER PACKAGE	SHARE OF DOCK-TO-DOOR
The warehouse	3.1	$75,060	$0.75	not included
Trucks to the hub	3.2: 64 × $640	$40,960	$0.41	16%
The hub	3.3	$30,288	$0.30	12%
Trucks to the depots	3.4: 64 × $384	$24,576	$0.25	10%
The depots	3.4: $0.30 × 100,000	$30,000	$0.30	12%
The vans	3.5: 500 × $264	$132,000	$1.32	51%
Everything		**$332,884**	**$3.33**	
Dock to door only (everything except the warehouse)	**$332,884 − $75,060**	**$257,824**	**$2.58**	**100%**

The two numbers everyone asks for. The first is the one just computed: *transportation cost per package*, which is every dollar spent from the warehouse dock to the customer's door, divided by the number of packages delivered. Today: $257,824 ÷ 100,000 = **$2.58**. The second is *the share of packages that arrived on the day we promised*. Today nothing went wrong, so it is **100%**. In Chapter 4 it will not be, and Chapter 4.7 shows exactly how it is computed when things go wrong.

The day in counts: how much of everything the network used

RESOURCE	COUNT	RESOURCE	COUNT
Pickers and packers per shift	58 + 58	Trailer trips to the hub	64
Warehouse paid hours	2,730	Trailer trips hub → depots	64
Hub paid hours	786	Hub doors in use per window	up to 12
Depot paid hours (all 17 depots)	about 940	Vans on the road	500
Van-driver hours	4,000	Delivery stops	80,000

3.7 The clock inside the warehouse: one truck, hour by hour

This is where the word *pile* earns its keep. Take the 9 PM truck. The orders that ride it arrive between 11 AM (right after the 3 PM truck's order deadline) and 5 PM (its own order deadline). The warehouse can pick-and-pack 5,000 packages an hour (58 packers a shift, minus breaks and supervisors, is about 50 people × 100 an hour). Everything must be packed by 7 PM. Here is the pile, hour by hour, on a normal day.

The 9 PM truck's work on a normal day: 25,000 packages

HOUR	ORDERS ARRIVING	WAITING + ARRIVING	PACKED THIS HOUR (5,000 AT MOST)	PILE LEFT AT THE END OF THE HOUR	SPARE CAPACITY USED FOR LATER TRUCKS
11 AM – noon	2,500	2,500	2,500	0	2,500
noon – 1 PM	3,000	3,000	3,000	0	2,000
1 – 2 PM	4,000	4,000	4,000	0	1,000
2 – 3 PM	5,000	5,000	5,000	0	0
3 – 4 PM	5,500	5,500	5,000	500	0
4 – 5 PM	5,000	5,500	5,000	500	0
5 – 6 PM	0 (deadline passed)	500	500	0	4,500

Everything for the 9 PM truck is packed by 5:06 PM. The deadline was 7 PM, so there is 1 hour 54 minutes to spare. The biggest pile was 500 packages, which at 5,000 an hour is a six-minute wait. Maria's package arrived at 2:12 PM, into an hour with no pile, which is why it was packed in under an hour.

Two habits of reading this table. The pile at the end of the hour is the stock; the 5,000 is the flow out; orders arriving is the flow in. Pile tomorrow = pile today + in − out. That one line is the whole of what people mean by a stock-and-flow model. And the number to watch is not the pile but the *time left when the pile reaches zero*: 1 hour 54 minutes here. When that number shrinks week after week, trouble is coming before anyone has missed a truck. Chapter 4.5 shows the same table on a day when it goes negative.

IF SOMEONE ASKS YOU TO EXPLAIN THIS IN ONE BREATH — 93 WORDS

A hundred thousand packages a day takes about fifty-eight pickers and fifty-eight packers a shift, sixty-four trailers to the hub and sixty-four from it, twelve hub doors, and five hundred vans. The warehouse costs seventy-five cents a package. From the dock to the door it is two dollars fifty-eight: sixteen percent first truck, twelve hub, ten second truck, twelve depots, and half of it vans. Inside the warehouse I watch one thing per truck: how much time is left when the pile for that truck hits zero. Today it was nearly two hours.

Chapter 4 — Now let things go wrong, one at a time

Each section adds exactly one problem to the happy-path day, works out what it costs and whom it hurts, lists the choices, and ends with the measurement you would put on a page so you see the problem before the customer does. The measurements are collected and defined properly in Chapter 5.

4.1 The truck is late

What goes wrong. The drive from the warehouse to the hub "usually" takes 4 hours. Trucks are not usual. Over the last 100 trips: half took 4 hours or less, 90 took 4 hours 27 minutes or less, 99 took 4 hours 49 minutes or less. The hub window is 1 AM to 4 AM. To clear the window, a trailer needs 2 hours to unload and about 25 minutes to sort and bag its last packages, so it must arrive by 4:00 − 2:25 = **1:36 AM**. A truck that left at 9:00 PM has 4 hours 36 minutes to make that.

How often, and what it costs

STEP	QUESTION	ARITHMETIC	ANSWER
1	How many trips take longer than 4 hours 36 minutes?	From the 100-trip record: 4 hours 36 minutes sits between the 90-in-100 mark (4:27) and the 99-in-100 mark (4:49). The curve gives about 4 in 100.	About 4% of trucks
2	How many late trucks a day?	64 trucks × 4%	About 2.8, call it 3 a day
3	What happens to their packages?	3 × 1,600 = 4,800 packages are still being unloaded when the window closes. They wait for the 7 AM window, leave the hub at 10 AM, reach the depot at 12:30 PM. The 9 AM vans are gone.	4,800 packages a day miss the morning vans
4	Cheapest rescue: the afternoon gig drivers	Gig drivers cost $1.90 a package; a regular van is $1.32. Extra: $0.58 × 4,800	$2,784 a day, and the customers still get it Tuesday
5	If there is no afternoon wave	4,800 packages wait for Wednesday	4,800 broken promises a day, 4.8% of the network

The three fixes, priced

FIX	ARITHMETIC	RESULT AND COST
Leave 15 minutes earlier (8:45 PM)	Now the truck has 4 hours 51 minutes. From the record, about 1 in 100 trips is that slow.	Late trucks fall from 3 a day to under 1. Cost: every customer's order deadline on this route moves from 5:00 PM to 4:45 PM; some of those orders are lost or go a day slower.
Put 3 unloaders on each door instead of 2	1,600 ÷ (3 × 400) = 1 hour 20 minutes to unload. The trailer can now arrive as late as 4:00 − 1:20 − 0:25 = 2:15 AM, which is 5 hours 15 minutes after a 9 PM departure. Almost no trip is that slow.	Late trucks fall to about zero. Cost: none in hours (the same 2 people-hours-per-400-packages, just three people for a shorter time), but the hub must staff more people at the start of the window and fewer later. **This is the fix to choose.**
Add 2 more hub doors	More doors do not make an individual truck unload faster. They help a different problem (bunching, below).	Wrong fix for this problem. Right fix for the next one.

The bunching problem, which looks the same and is not. Suppose every truck is on time but they arrive in clumps: 6 at 1:00, 6 at 1:30, 4 at 2:00. Twelve doors. The first twelve trucks take the twelve doors until 3:00 and 3:30. The last four wait for a door until 3:00, and finish unloading at 5:00, an hour after the window closed. Four trailers, 6,400 packages, roll to the next window, even though no truck was late. The averages hide it completely: door use was 16 trailers × 2 hours ÷ (12 doors × 3 hours) = 89%, which sounds fine. The fix here is either the 3-person crews (the first twelve finish by 2:20 and 2:50, the last four start at 2:20 and finish by 3:40), or spacing the departures at the warehouse, or, if it happens every night, two more doors.

Measurements this creates. (a) *Trucks arriving on time*: of the trucks due in a window, the share that arrived before the latest useful time (1:36 AM with 2-person crews). (b) *Trucks leaving on time*: of the trucks scheduled to leave the warehouse, the share that left within 15 minutes of schedule. Keep them separate: a truck that left on time and arrived late is the road's problem; one that left late is the warehouse's. (c) *Hub window finished*: of the packages that arrived for a window, the share bagged before it closed.

4.2 The trailer is half empty

What goes wrong. The truck costs $640 whether it carries 2,000 packages or 200. Fewer packages aboard means each one pays more.

The same truck, different loads

PACKAGES ABOARD	ARITHMETIC	COST PER PACKAGE	COMPARED WITH THE PLAN
2,000 (completely full)	$640 ÷ 2,000	$0.32	8 cents cheaper
1,600 (the plan, 80%)	$640 ÷ 1,600	$0.40	—
1,200 (60%)	$640 ÷ 1,200	$0.53	13 cents dearer
800 (40%)	$640 ÷ 800	$0.80	double

A small drop, across the whole day

STEP	QUESTION	ARITHMETIC	ANSWER

1	If the average trailer carried 1,500 instead of 1,600, how many trailers a day?	100,000 ÷ 1,500	67 instead of 64
2	Extra cost?	3 extra trailers × $640	$1,920 a day
3	Over a year?	$1,920 × 360 days	About $690,000 for one warehouse, one route, from a drop nobody would notice looking at a trailer

Why not plan for 2,000, then? Because the number of packages for a given truck is not the same every day. Say it swings by about 10% around the plan. If we planned 1,800 per trailer, a 10% swing is 180 packages; a day that is 200 over the plan happens about 13% of the time, and on those days packages are left on the dock when the truck leaves. If we plan 1,600, the swing is 160 and we would need to be 400 over, which is two and a half swings; that happens well under 1% of the time. Eighty percent is the balance between a truck that is cheap and a dock that is empty when the truck leaves. The right percentage is different on every route and is one of the numbers this job owns.

The trade this creates. At 8:55 PM a trailer has 1,100 packages. Do you hold it twenty minutes to get it to 1,400? The truck gets cheaper ($640 ÷ 1,100 = $0.58 versus $640 ÷ 1,400 = $0.46), but it eats into the spare 36 minutes at the hub, and 4.1 just showed what that costs. The answer depends on the spare time on that route tonight, and that is exactly the kind of rule Chapter 7's dashboard has to show the dock lead in real time.

Measurement this creates. *Trailer fullness*: packages aboard when the doors close ÷ 2,000, averaged over the day's trailers on a route. Plan 80%. Shown next to it: how many trailers went out under 60%.

That measurement uses the package count. It cannot see air inside a gaylord that was closed early. Section 8.10 is the camera that measures that air, and the arithmetic of what the air is worth.

4.3 More packages than vans

What goes wrong. Depot 7 planned for 6,000 packages and 30 vans. This morning 6,600 arrived.

The three choices at 8 AM

STEP	QUESTION	ARITHMETIC	ANSWER
1	How many vans are needed?	6,600 ÷ 200	33. There are 30.
2	Choice A: find 3 more vans and drivers by 9 AM	3 × $264	$792. Usually impossible on an hour's notice.
3	Choice B: send the extra 600 with afternoon gig drivers	600 × $1.90	$1,140, which is $348 more than regular vans would have cost. Customers still get it today.
4	Choice C: leave 600 for tomorrow	600 ÷ 6,600	9% of the depot's customers get it a day late. Cheapest in dollars, dearest in promises.
5	Choice D, the one that looks free and is not: squeeze 220 into each van	30 × 220 = 6,600; 220 ÷ 1.25 = 176 stops ÷ 20 an hour = 8.8 hours of stops	Fits inside the 11-hour window with 2.2 hours for breaks and the drive back. Works today. Does not work at 260 (10.4 hours).

The opposite mistake. A depot that is scared of this will plan too many vans. If Depot 7 sends 32 vans for 6,000 packages, each carries 6,000 ÷ 32 = 187.5, and the cost per package is $264 ÷ 187.5 = $1.41 instead of $1.32. Nine cents × 6,000 = $528 a day, at one depot, every day. Across 17 depots that is about $9,000 a day. This is why "packages per van" is the most expensive number in the network: too few vans breaks promises; too many burns money quietly.

Measurements this creates. *Left behind at the depot*: packages that arrived for the morning vans and did not go out on them ÷ packages that arrived. Plan under 0.5%. And *packages per van*: packages sent ÷ vans sent. Plan 200.

4.4 A package goes to the wrong building

What goes wrong. The hub's sorting machine drops a package down the chute for Depot 12 instead of Depot 7, or a bag is labeled wrong, or the table that says which streets belong to which depot has a mistake. The package rides to the wrong depot, is discovered at the scan there, and has to be sent back through the hub.

Three in a thousand

STEP	QUESTION	ARITHMETIC	ANSWER
1	How many a day?	100,000 × 3 ÷ 1,000	300 packages
2	What does each cost to fix?	A second sort at the wrong depot ($0.30) + a ride back and out again ($0.24 + $0.24) + a share of the customer-service calls and refunds	About $2.00 each; $600 a day
3	How many arrive late?	Most miss the day. Say 7 in 10: 300 × 0.7	210 broken promises a day, 0.2% of the network, from one building's accuracy

Small in dollars, large in promises. And it is usually not the machine's fault: the most common cause is a change to the street-to-depot table that someone made for a good reason (a new depot opened; a route was redrawn) and that left two streets pointing at the old depot. Chapter 6's example week has exactly that.

Measurement this creates. *Sent to the wrong building*: packages scanned at a depot other than the one on the label ÷ packages sorted. Plan under 0.3%.

4.5 A surge of orders

What goes wrong. A promotion. Orders for the 9 PM truck come in 60% heavier than normal: 40,000 instead of 25,000. Packing capacity is still 5,000 an hour. Same table as 3.7.

The 9 PM truck's work on the surge day: 40,000 packages

HOUR	ORDERS ARRIVING	WAITING + ARRIVING	PACKED THIS HOUR (5,000 AT MOST)	PILE LEFT AT THE END OF THE HOUR	WAIT FOR A PACKAGE ARRIVING NOW
11 AM – noon	4,000	4,000	4,000	0	none
noon – 1 PM	4,800	4,800	4,800	0	none
1 – 2 PM	6,400	6,400	5,000	1,400	1,400 ÷ 5,000 = 17 min
2 – 3 PM	8,000	9,400	5,000	4,400	53 min
3 – 4 PM	8,800	13,200	5,000	8,200	1 h 38 min
4 – 5 PM	8,000	16,200	5,000	11,200	2 h 14 min
5 – 6 PM	0 (deadline)	11,200	5,000	6,200	
6 – 7 PM	0	6,200	5,000	1,200	
7 – 8 PM	0	1,200	1,200	0	

Packing finishes at 7:14 PM. The deadline was 7:00. 1,200 packages are packed too late to be labeled, sorted, and loaded by 9:00. They miss the truck. (The total work was 40,000 ÷ 5,000 = 8 hours; the window from 11 AM to 7 PM is 8 hours; there was no slack at all, and the late-afternoon bunching made it worse.)

The choices, priced

CHOICE	ARITHMETIC	RESULT
Add packing capacity from 10 AM	Need 1,200 more packed by 7 PM; 1,200 ÷ 6 hours = 200 an hour; that is 2 more packers (100 an hour each) and 2 more pickers (1.6 items × 200 = 320 items an hour ÷ 160)	4 extra people for one shift, about $350. Everyone makes the truck. But you must know by 10 AM that the surge is coming: that is what a forecast is for.
Move the customer deadline to 4:45 PM for today	Roughly 15 minutes of orders at 8,000 an hour = 2,000 orders go to the 3 AM truck instead	Those customers see Wednesday instead of Tuesday on the website. Some do not buy.
Let the 1,200 miss the truck	They ride the 3 AM truck, reach the hub at 7 AM, the depot at 12:30 PM, and go out with the afternoon gig drivers at $0.58 extra = $696, or wait until Wednesday	Cheap if the afternoon wave exists; 1,200 broken promises if not.

Measurement this creates. *Packing finished ahead of the truck*: the clock time the last package for a truck was packed, subtracted from the deadline, in hours. Normal day 1.9 hours. Surge day −0.24 hours. This is the earliest warning the warehouse has: it moves hours before a truck is actually missed, and days before customers notice.

4.6 Not every package takes the same road

What changes. So far every package went warehouse → hub → depot. Real networks have three roads, and choosing between them is the core of this job.

Road 1 versus Road 2: through the hub, or straight to the depot

STEP	QUESTION	ARITHMETIC	ANSWER
1	What does the middle of the journey cost through the hub?	First truck $0.40 + hub $0.30 + second truck $0.24	$0.94 per package (the depot and van are the same ei

2	Depot 3 is 90 miles from the warehouse and gets 1,500 of our packages a day. A truck straight there?	90 miles × $3.20 = $288; ÷ 1,500	$0.19 per
3	Saving per package?	$0.94 − $0.19	$0.75; on 1,500 a day, $1,125 a day, about $400,000 a year for c
4	Time saved?	Through the hub: 4 h drive + 3 h window + 2.5 h drive = 9.5 hours. Direct: 2 hours.	7.5 hours. Either Depot 3's customers can order 7.5 hours later for the same promise, or get a faster
5	When does a direct truck lose? Depot 9 gets only 200 of our packages a day.	$288 ÷ 200	$1.44 per package, worse than $0.94 through the hub. The hub exists to combine small flows into f
6	The break-even	$288 ÷ $0.94	306 packages a day. Above that, send a direct truck; below it, use the hub. (Longer distances and lighter trucks move this number; the method is th

Road 3: hand it to an outside carrier

STEP	QUESTION	ARITHMETIC	ANSWER
1	Some addresses are too far or too few for our vans. The postal service or another carrier takes the package at our dock for $4.50 dock-to-door.	$4.50 against $2.58 through our own network	$1.92 more per package. Worth it only where our own road does not exist or would be thinner than a carrier's.
2	Suppose today 70,000 go through the hub, 15,000 go direct, 15,000 go to a carrier. What is the day's cost per package?	Hub road $2.58 (we'll use the planned $2.56 for the arithmetic below); direct road $0.19 + $0.30 + $1.32 = $1.81; carrier $4.50. Weighted: 0.70 × $2.56 + 0.15 × $1.81 + 0.15 × $4.50 = $1.792 + $0.272 + $0.675	$2.74 per package
3	Now 2,000 packages a day that used to go through the hub go to the carrier instead (a new area, or a van shortage). Nothing else changes.	2,000 × ($4.50 − $2.56)	$3,880 a day more, which is about 4 cents on every one of the 100,000 packages, even though not a single road got more expensive.

That last row is the most misunderstood thing on a weekly cost report. The cost per package can go up because *the mix of roads changed*, with every road costing exactly what it did before. Any honest report must split a change in cost per package into two parts: the part from each road's own cost changing (rate), and the part from packages moving between roads (mix). Chapter 6.2 does this with real numbers.

Measurement this creates. *Share of packages on each road*: hub, direct, carrier, each ÷ total packages. Plan 70 / 15 / 15 in our example.

4.7 Putting the problems together: how many packages arrive on the promised day

Each problem above sends some packages off the happy path. Some of those still arrive on time (the afternoon gig drivers rescue them); some do not. The share that arrive on the promised day is simply 100% minus the sum of the failures, each weighted by how often it actually loses the day.

A normal week in our network, with all four problems happening at their normal rates

PROBLEM	HOW MANY PACKAGES IT HITS	HOW OFTEN THAT LOSES THE DAY	SHARE OF ALL PACKAGES ARRIVING LATE	WHY THAT FRACTION
Missed the warehouse truck (4.5)	0.8%	4 in 10	0.8 × 0.4 = 0.32%	The next truck often still makes the depot deadline
Truck late to the hub or depot (4.1)	1.2%	5 in 10	1.2 × 0.5 = 0.60%	Half are rescued by the afternoon gig wave
Sent to the wrong building (4.4)	0.3%	7 in 10	0.3 × 0.7 = 0.21%	Two extra truck rides; usually a day
Left behind at the depot (4.3)	0.6%	10 in 10	0.6 × 1.0 = 0.60%	By definition it waits for tomorrow's vans
Total arriving late (own network)			**1.73%**	
Outside carrier late on its 15% of packages	15% × 3% late	—	0.45%	Their performance, our promise
Blended: 85% own network + 15% carrier			**0.85 × 1.73% + 0.45% = 1.92%**	
Arrived on the promised day			**98.1%**	**Out of 700,000 packages in a week, about 13,400 late**

Read the "share arriving late" column and you have the priority list for speed: late trucks and packages left behind at depots are each 0.6%; everything else is smaller. Read Chapter 4.2 and 4.3 again and you have the priority list for cost: packages per van, then which road, then trailer fullness. The two lists are different, and the job is to hold both at once.

IF SOMEONE ASKS YOU TO EXPLAIN THIS IN ONE BREATH — 118 WORDS

Four things go wrong on a normal day. A truck is late, so its sixteen hundred packages wait three hours for the next sorting window; the cheap fix is three unloaders per door, not an earlier departure. A trailer leaves light, which quietly costs seven hundred thousand a year per route for a drop nobody sees. A depot gets more packages than it has vans, and the cheapest choice in dollars, leaving them for tomorrow, is the dearest in promises. And one package in three hundred goes to the wrong building. Add up each one times how often it really costs a day, and you get the share delivered on the promised day: ninety-eight percent in our network.

Chapter 5 — The measurements, explained from the example

A measurement is a question with a fraction attached. The top of the fraction is a count of something; the bottom is the count of everything it could have been; the result is checked every week against a plan. Every measurement below was already met in Chapter 4 as a problem. Nothing new is introduced here; it is only collected and made precise.

5.1 The two numbers at the top, word by word

Transportation cost per package

PART OF THE DEFINITION	PLAIN MEANING	OUR EXAMPLE
Top of the fraction	Every dollar spent after the package leaves the warehouse dock: the trucks (fuel, drivers, tractors, trailers), the hub (people, rent, machine), the second trucks, the depots (people, rent), the vans (drivers, vehicles, fuel), the gig drivers, the outside carriers, and the cost of fixing mistakes (wrong-building re-sorts). *Not* included: the warehouse's own picking and packing, which is reported on its own line so the two cannot be confused.	$257,824 on the happy-path day
Bottom of the fraction	Packages *delivered* (or attempted) in the period, counted at the depot that delivered them. Not packages shipped: a package shipped Saturday night and delivered Monday would otherwise count in one week's cost and the next week's packages.	100,000
Result	Top ÷ bottom	$2.58
How often	Weekly in dollars (invoices from carriers and van companies arrive late, so a daily dollar figure would be a guess). The operational pieces underneath it (fullness, packages per van) are watched daily.	Week 45: $2.86 (Chapter 6)
Split by	Region; road (hub, direct, carrier); and type of package (small, standard, large), because each has its own plan.	Hub road $2.56; direct $1.81; carrier $4.50
Against what	A plan that is set per road, not one blended number, so that a change in mix is visible instead of hiding inside a rate.	Plan $2.74 blended (70/15/15 mix)

Share of packages that arrived on the promised day

PART OF THE DEFINITION	PLAIN MEANING	OUR EXAMPLE
Top of the fraction	Packages where the driver made a delivery attempt on or before the date the website promised. An attempt counts even if nobody was home, because the network did its part.	686,560 in a 700,000 week
Bottom of the fraction	Packages with a promised date in the period.	700,000
Result	Top ÷ bottom	98.1%
How often	Daily (the signal is immediate) and weekly for the trend.	
Split by	Cause: missed truck, late truck, wrong building, left behind, carrier. This split is what makes the number actionable, because each cause has an owner.	0.32 / 0.60 / 0.21 / 0.60 / 0.45 points
Note	Amazon has publicly used the name *delivery estimate accuracy* for this idea, and has not published its rate. Our 98.1% is the made-up network's number.	

5.2 The twelve measurements: the question each answers, the fraction, our number

MEASUREMENT	THE QUESTION IT ANSWERS	THE FRACTION	OUR EXAMPLE	PLAN	WHO OWNS IT	WHEN IT IS BAD, WHAT YOU DO
Results (what the customer and the finance team feel)						
Transportation cost per package	What did it cost us to move a package from the dock to the door?	Dock-to-door dollars ÷ packages delivered	$2.58	$2.74 blended	This role	Split the change into mix and rate (6.2); hand each piece to its owner
Arrived on the promised day	Did we keep the promise the website made?	Packages attempted by the promised date ÷ packages promised	98.1%	≥ 97.5%	Regional operations	Read the cause split; the biggest cause gets the first meeting
Packages against plan	Did the volume we planned for show up?	Packages delivered ÷ packages planned	100,000 ÷ 100,000	within 3%	Planning	Outside 3%, the plan was wrong, not the network; fix the forecast before blaming a building
Dollars against plan	How many dollars over or under were we this week?	(actual cost per package − planned) × packages	$0 on the happy day	$0	Finance	This is the number senior leaders ask about; the bridge in 6.2 is the answer
Drivers (the denominators the results are made of)						

Packages per van	How many packages shared each van's $264?	Packages sent out ÷ vans sent out	200	200	Depot manager and van company	Re-plan vans to the package forecast, not to yesterday's van count (4.3)
Trailer fullness	How much of each truck's $640 was carrying packages?	Packages aboard at door-close ÷ 2,000, averaged	78% (1,563 ÷ 2,000)	80%	Transportation	Find the route and the leave time; remove a departure or move volume onto it (4.2)
Share on each road	How many packages went through the hub, direct, or to a carrier?	Packages on each road ÷ all packages	70 / 15 / 15	same	This role	A shift toward the carrier is the most expensive thing that can happen without anyone making a mistake (4.6)
Hub cost per package	Is the hub spreading its rent over enough packages?	Hub dollars ÷ packages sorted	$0.30	$0.30 at 100,000	Hub manager	If volume is down, this rises by itself; it is a planning signal, not an operating one (3.3)
Daily operating signals (what a building can change tomorrow)						
Packing finished ahead of the truck	How close did we come to missing the truck?	Truck deadline − time the last package was packed, in hours	1.9 hours	≥ 1.5	Warehouse	The earliest warning there is; add people to the next shift before the truck is missed (4.5)
Trucks arriving on time	Did the trucks reach the hub or depot in time to be unloaded before the deadline?	Trucks arrived before the latest useful time ÷ trucks due	61 ÷ 64 = 95%	≥ 97%	Transportation (the road) and the hub (the doors)	If the trucks left on time, fix the road plan or the crew size; if they left late, go to the warehouse (4.1)
Left behind at the depot	How many packages reached the depot and did not go out that day?	Packages not sent out on the planned wave ÷ packages received for it	0.6%	≤ 0.5%	Depot manager	Count vans against the package forecast; arrange gig capacity before the wave, not after (4.3)
Sent to the wrong building	How many packages did the hub mis-route?	Packages scanned at a depot other than the one on the label ÷ packages sorted	0.3%	≤ 0.3%	Hub manager	Check the last change to the street-to-depot table first; then the label scanner; then the chutes (4.4)

Three rules for any measurement on the page. (1) The bottom of the fraction is printed in the column header, every week, so two people cannot argue about different denominators. (2) The dollars must add up to what finance paid, within 1%, and the package counts must match the delivery system within half a percent, or the page is not published. (3) Compare like with like: this week against last week on the same depots and the same roads; a plan made in January against the January forecast, not against what we know now.

5.3 Which ones matter most, with the money behind the ranking

"Matters most" has two meanings: dollars per week for a normal-sized move, and broken promises per week for a normal-sized move. Both are computed on the happy-path network (100,000 packages a day, 700,000 a week).

For cost: dollars in a week from a move the network actually makes in a normal week

RANK	MEASUREMENT	A NORMAL WEEK'S MOVE	ARITHMETIC	DOLLARS A WEEK
1	Packages per van	200 → 190	$264 ÷ 190 − $264 ÷ 200 = $0.07 more per package × 100,000 a day × 7	about $48,600
2	Share on each road	2 points move from hub to carrier (2,000 a day)	2,000 × ($4.50 − $2.56) = $3,880 a day × 7	about $27,200
3	Trailer fullness	80% → 77% (3 points)	Per point, both trucks: ($640 + $384) ÷ 1,580 − ($640 + $384) ÷ 1,600 = $0.008 × 100,000 = $810 a day; × 3 points × 7	about $17,000

4	Hub cost per package	Volume 100,000 → 92,000 (8% short of plan)	$13,000 ÷ 92,000 − $13,000 ÷ 100,000 = $0.011 × 92,000 = $1,040 a day × 7	about $7,300, and it is a planning miss, not an operating one
5	Sent to the wrong building	0.3% → 0.45%	150 more a day × $2 × 7	about $2,100
6	Left behind at the depot	0.5% → 0.7%	200 more a day × $0.58 (if rescued by gig drivers) × 7	about $800 in dollars; see the next table for what it really costs

For promises: broken promises in a week from a normal-sized move

RANK	MEASUREMENT	A NORMAL WEEK'S MOVE	ARITHMETIC	LATE PACKAGES A WEEK
1	Left behind at the depot	+0.2 points	0.2% × 700,000 × 10 in 10 lose the day	1,400
2	Trucks arriving on time	−1 point (one more truck in 100 late)	1% of packages × 700,000 × 5 in 10	3,500 per point; a normal week moves it about 0.3 points, so about 1,050
3	Trucks leaving on time (missed truck)	−0.5 points	0.5% × 700,000 × 4 in 10	1,400
4	Sent to the wrong building	+0.15 points	0.15% × 700,000 × 7 in 10	735
5	Share to carrier	+2 points	2% × 700,000 × 3% late	420 (but rank 2 for cost)

So, in plain words. For money, watch packages per van first, then which road the packages take, then how full the trailers are. For promises, watch packages left behind at depots first, then trucks arriving on time, then trucks leaving on time. Two signals carry no dollars at all and are watched every morning anyway, because they move first: *packing finished ahead of the truck* and *hub door use in the window*. They are the smoke before the fire.

IF SOMEONE ASKS YOU TO EXPLAIN THIS IN ONE BREATH — 112 WORDS

Two numbers sit at the top: what it cost to get a package from the dock to the door, and how many arrived on the day we promised. Under them are the denominators that make the cost, packages per van, how full the trailers were, and which road the packages took. Under those are the daily operating signals a building can move tomorrow: how much time was left when packing finished, trucks arriving on time, packages left behind, packages sent to the wrong building. Ten packages a van is worth fifty thousand dollars a week in our example. Two tenths of a point of packages left behind is fourteen hundred broken promises.

Chapter 6 — The weekly scorecard

One page, every Monday morning, for the week that ended Saturday. It shows the measurements from Chapter 5 against last week and against the plan, and for every number that moved, one line saying why and whose it is. The example week is week 45, a promotion week: 8% more packages than the week before, and cost per package up twelve cents.

6.1 The page itself: this week, last week, the plan

Colors follow one rule: green within 2% of plan, yellow within 5%, red beyond 5% (for percentages, the same idea in points). A red or yellow cell must have a cause and an owner on the same line; "volume" and "peak" are not causes.

MEASUREMENT	PLAN	WK 42	WK 43	WK 44	WK 45	CHANGE	AGAINST PLAN	CAUSE, OWNER
Packages delivered (thousands)	728	693	697	700	756	+8%	+4%	Promotion week; inside the 3% band once the re-forecast is used. Planning.
Transportation cost per package	$2.74	2.73	2.75	2.74	2.86	+12¢	+4.4%	Vans +5.7¢, carrier share +4.2¢, trailer fullness +2.3¢, hub rent −0.7¢ (see 6.2). Three owners below.
Dollars against plan (thousands)	0	−7	+7	0	+89			$0.118 × 756,000 packages.
Cost per package, hub road	$2.56	2.56	2.57	2.56	2.66	+10¢	+3.8%	Fullness 76% on hub trucks from Tuesday; packages per van. Transportation; depots.
Cost per package, direct road	$1.81	1.81	1.82	1.81	1.88	+7¢	+3.8%	Packages per van only.
Cost per package, carrier	$4.50	4.50	4.50	4.50	4.50	0	0	Contract rate. Week 42's late invoice posted this week: +$14,000, shown separately, not in the rate.
Share of packages to carrier	15.0%	14.8	15.1	15.0	17.0	+2 pts	+2 pts	Promotion overflow from two depots handed to the carrier Wed–Sat. This role: price an afternoon own-van wave instead, by Friday.
Trailer fullness, hub trucks	80%	80.1	79.8	80.0	76.0	−4 pts	−4 pts	A 17th trailer was added to each leave time on Tuesday for the promotion and not removed when Thursday's volume fell. Transportation: remove it or fill it, by Wednesday.
Packages per van	200	199	201	200	190	−5%	−5%	Depots added vans faster than packages: 104% of planned vans on 108% of planned packages. Depot managers: plan vans from the package forecast. Thursday.
Hub packages a day (thousands)	104	99	100	100	108	+8%	+4%	Rent spread over more packages: −0.7¢. Good. No action.
Sent to the wrong building	0.30%	0.29	0.31	0.30	0.45	+0.15 pts	+0.15 pts	Street-to-depot table was changed for the new direct route to Depot 3 and left two streets on the old depot. Hub: fixed Thursday; confirm under 0.30% next week.
Arrived on the promised day	≥ 97.5%	98.2	98.1	98.1	97.9	−0.2 pts	+0.4 pts	Above plan, but down: wrong building −0.1, carrier share −0.07, left behind −0.05.
Left behind at the depot	≤ 0.5%	0.4	0.5	0.55	0.6	+0.05	+0.1	Third week rising in a row: deep dive opened. Two depots short of vans on Mondays. Depot managers.
Trucks arriving on time	≥ 97%	96.9	97.1	96.3	96.1	−0.2	−0.9	Door bunching at the hub's 1 AM window Tuesday and Thursday (4 trailers waited). Hub: 3-person crews on trial from Monday.
Trucks leaving on time	≥ 98%	98.3	98.1	98.1	98.0	−0.1	0	—
Packing finished ahead of the truck (hours)	≥ 1.5	1.9	1.8	1.7	1.3	−0.4	−0.2	Early warning. Promotion volume at the 9 PM truck. Warehouse: watch daily; move people from the 3 AM truck's shift if it falls under 1.0.
Open actions. (1) Price an afternoon own-van wave against the carrier for promotion overflow — this role — Friday. (2) Remove or fill the 17th trailer — transportation — Wednesday. (3) Van plan from package forecast at the two depots — depot managers — Thursday. (4) Confirm wrong-building rate back under 0.30% — hub — next week. (5) Left-behind deep dive — depots — report next Monday. (6) 3-person crew trial, 1 AM window — hub — report next Monday. Carried from week 44: Monday van capacity at Depots 7 and 12 — open.								

6.2 "Why did cost go up twelve cents?" answered in plain words, with the arithmetic

This is the question a senior leader will ask, and the one most reports answer badly. The honest answer has two parts, because cost per package can rise for two completely different reasons: packages moved onto a more expensive road (*mix*), or a road itself got more expensive (*rate*). Here is week 44 to week 45, worked.

Step 1 — What we know about the two weeks

ROAD	WK 44 SHARE	WK 44 COST	WK 45 SHARE	WK 45 COST	WHAT CHANGED ON THAT ROAD
Through the hub	70.0%	$2.56	68.4%	$2.66	Trailer fullness 80% → 76%; packages per van 200 → 190; wrong building 0.30% → 0.45%; hub volume 100,000 → 108,000 a day
Direct to depot	15.0%	$1.81	14.6%	$1.88	Packages per van 200 → 190

Outside carrier	15.0%	$4.50	17.0%	$4.50	Share up 2 points; rate unchanged
All packages	**100%**	**$2.74**	**100%**	**$2.86**	**+$0.118, which × 756,000 packages = $89,200 for the week**

Step 2 — How much is because packages moved between roads? (mix)

ROAD	CHANGE IN SHARE × LAST WEEK'S COST ON THAT ROAD	CENTS PER PACKAGE
Through the hub	(68.4% − 70.0%) × $2.56 = −1.6% × $2.56	−4.1¢
Direct	(14.6% − 15.0%) × $1.81 = −0.4% × $1.81	−0.7¢
Carrier	(17.0% − 15.0%) × $4.50 = +2.0% × $4.50	+9.0¢
Mix, total	**Two percent of packages moved to a road that costs $1.94 more**	**+4.2¢**

Step 3 — How much is because a road itself got dearer? (rate)

ROAD	THIS WEEK'S SHARE × CHANGE IN THAT ROAD'S COST	CENTS PER PACKAGE
Through the hub	68.4% × ($2.66 − $2.56) = 68.4% × $0.096	+6.6¢
Direct	14.6% × ($1.88 − $1.81) = 14.6% × $0.069	+1.0¢
Carrier	17.0% × $0	0
Rate, total		**+7.6¢**
Check: mix + rate	**4.2 + 7.6**	**11.8¢ — matches the top line**

Step 4 — Why did the hub road get 9.6 cents dearer? Open it up

CAUSE	ARITHMETIC (PER PACKAGE ON THE HUB ROAD)	CENTS
Vans carried 190 instead of 200	$264 ÷ 190 − $264 ÷ 200 = $1.389 − $1.320	+6.9¢
Trailers 76% full instead of 80%	Both trucks: ($640 + $384) ÷ 1,520 − ($640 + $384) ÷ 1,600 = $0.674 − $0.640	+3.4¢
Wrong-building rate 0.45% instead of 0.30%	0.15% × $2.00	+0.3¢
Hub rent spread over 108,000 instead of 100,000	$13,000 ÷ 108,000 − $13,000 ÷ 100,000 = $0.120 − $0.130	−1.0¢
Total	**6.9 + 3.4 + 0.3 − 1.0**	**+9.6¢ — matches Step 3**

Step 5 — The answer, in one paragraph, with every cent accounted for

IN PLAIN WORDS	SPREAD OVER ALL 756,000 PACKAGES	CENTS PER PACKAGE
Vans went out with 190 packages instead of 200 (on the hub and direct roads, 83% of packages)	83% × 6.9¢	+5.7¢
Two percent more packages were handed to the outside carrier	Step 2	+4.2¢
Trailers to and from the hub were 76% full instead of 80% (68.4% of packages)	68.4% × 3.4¢	+2.3¢
More packages were sent to the wrong building	68.4% × 0.3¢	+0.2¢
The hub's rent was spread over more packages (good news)	68.4% × −1.0¢	−0.7¢
Total		**+11.8¢ ≈ 12 cents; $89,200 for the week**

What the bridge says to do. Half the increase is vans, and it belongs to the depots: they added vans ahead of packages. A third is the carrier share, and it belongs to this role: the promotion overflow should have gone to an own-van afternoon wave if that is cheaper than $4.50, and that calculation is the first action on the list. A fifth is trailer fullness, and it belongs to transportation: a trailer added for Tuesday's surge was still running on Saturday. The wrong-building line is small in cents and matters on the promise page, not this one. The hub's rent line is good news and gets one clause.

6.3 How the weekly meeting runs

MINUTES	WHAT HAPPENS	RULE
0–5	Everyone reads the page in silence.	Nobody presents numbers. The page has to stand on its own.
5–10	Did the numbers reconcile? Any definition change?	If the dollars did not tie to finance within 1%, the affected lines are marked "not reconciled" and nobody discusses causes on them. A changed definition is shown old and new, side by side, for four weeks.
10–30	Red lines, largest dollars first, then largest promise effect.	The owner speaks. Seven words in a fixed order: measurement, plan, actual, cause, action, owner, date. "Volume" is not a cause; "monitor" is not an action.
30–38	Yellow lines and three-week trends.	Three weeks moving the same way opens a deep dive even if no week was red. (Left-behind, this week.)
38–43	Open actions from earlier weeks.	An action closes only when its measurement moved. Otherwise it gets a new date and a reason.
43–45	The week in one sentence.	Said by the chair, written at the top of next week's page: "Week 45: cost up twelve cents on eight percent more packages; half of it vans added ahead of volume, a third carrier overflow; promises held at 97.9."

IF SOMEONE ASKS YOU TO EXPLAIN THIS IN ONE BREATH — 118 WORDS

The scorecard is one page every Monday: this week, last week, the plan, and a color. Under the cost line there is always a bridge, because cost per package can go up for two different reasons: packages moved to a dearer road, or a road got dearer. This week it was twelve cents: five point seven from vans carrying ten fewer packages, four point two from two percent more packages going to the carrier, two point three from trailers four points emptier, and the hub's rent gave back seven tenths. Every cent has an owner and a date. We read it in silence, take the red lines by size, and close an action only when the number moves.

Chapter 7 — How this role would build the dashboard

A dashboard is the scorecard made live: the same measurements, the same definitions, refreshed as the scans come in, with the ability to click any number and see the packages, trailers, or vans behind it. Here is how someone in this role would build one, in the order they would do it, with the reasoning.

7.1 Start with the people and their questions, not with charts

The first mistake is to open a charting tool. The first step is a table of who will look at the screen, how often, what decision they are about to make, and what question answers it. Everything else follows from this table; a tile that does not answer a question on it does not get built.

WHO	HOW OFTEN	THE DECISION THEY ARE ABOUT TO MAKE	THE QUESTION THE SCREEN MUST ANSWER	WHICH SCREEN
Warehouse shift lead	Every 30 minutes	Move people between picking, packing, and loading; call in extra hands for the next truck	For each of the next two trucks: how big is the pile, and how much time is left when it reaches zero?	Today
Dock lead at the warehouse	Every 15 minutes near a leave time	Close the trailer now, or hold it 20 minutes to fill it	How full is each trailer, how many packages are still coming for it, and how much spare time does this route have at the hub tonight?	Today
Hub window lead	Continuously during a window	Put a third unloader on a door; open another door; warn the depot a trailer will be late	Which trucks are on the road to me, when will each arrive, and will each be unloaded before the window closes?	Today
Depot manager	5 AM and 8 AM	How many vans to dispatch; whether to book afternoon gig drivers	How many packages have arrived or are about to, how many vans do I have, and how many packages would be left behind?	Today
Regional operations director	Once a day, 8 AM	Where to send attention today; which building manager to call	Yesterday, which buildings missed their plan on which measurement, and why?	Yesterday
This role	Daily and weekly	What to change in the network: a leave time, a route, a crew size, a road	Which measurement moved, is it one building or a pattern, and what is it costing per week?	All three
Transportation finance	Weekly	Confirm the dollars; update the plan of record	Do the dollars on the page tie to the ledger, and how much of the variance is mix versus rate?	This week
Vice president	Weekly, 5 minutes	Whether to intervene	Cost per package and promises kept, against plan, with the one-sentence reason	This week, top row only

Three things fall out of this table before any data is touched. The hourly users need *counts and clocks*, never dollars: dollars arrive weeks later on invoices, so an hourly dollar figure would be a guess that people would argue about. The daily user needs *causes*, not just numbers: a red cell with no cause is a phone call, not a decision. The weekly users need the *bridge* from Chapter 6.2 built in, not computed by hand on Sunday night.

7.2 Three screens: today, yesterday, this week

Screen 1 — Today (refreshed every 10 minutes; counts and clocks only)

```
TODAY · Warehouse A · Monday 3:40 PM                                  refreshed 3:38 PM · scans tying: OK

 THE NEXT TWO TRUCKS                                TRAILERS AT THE DOORS (9 PM leave time)

 ┌──────────────┬──────────┬──────────┬────────────┐  ┌──────┬─────────┬──────────┬────────────┬──────────────┐
 │ Leave time   │ Pile now │ Pack by  │ Time left  │  │ Door │ Route   │ Aboard   │ Still due  │ Spare at hub │
 ├──────────────┼──────────┼──────────┼────────────┤  ├──────┼─────────┼──────────┼────────────┼──────────────┤
 │ 9 PM  (hub)  │   8,200  │ 7:00 PM  │ -0.2 h  RED│  │ 14   │ Hub     │ 1,090    │    640     │ 36 min       │
 │ 3 AM  (hub)  │     900  │ 1:00 AM  │  5.8 h     │  │ 15   │ Hub     │   310    │ 1,300      │ 36 min       │
 └──────────────┴──────────┴──────────┴────────────┘  │ 22   │ Depot 3 │   880    │    420     │ 1 h 10 min   │
   Pile is 11,200 at 4 PM if orders keep this pace.  └──────┴─────────┴──────────┴────────────┴──────────────┘
   Suggested: +2 packers, +2 pickers now (200/hour).    Fullness so far today: 79%  (plan 80)

 TRUCKS ON THE ROAD TO THE HUB (7 AM window closed; 1 PM window)       LABELING MACHINES   this hour 6,900 / 7,500 = 92%
   Trailer 11-C  left 9:02 AM  expected 1:08 PM  latest useful 1:36 PM   ok                  — 93% is the knee. Watch.
   Trailer 11-D  left 9:20 AM  expected 1:41 PM  latest useful 1:36 PM   LATE 5 min → hub told: 3rd unloader on door 4
   ...

 DEPOT 7 · tomorrow morning                          WRONG BUILDING today: 9 of 31,400 sorted (0.03%)   ok
   Expected packages 6,400 · vans planned 30 → 200/van would leave 400 behind · gig blocks booked: 0  → ACTION
```

Every number on this screen is a count or a clock from scans that happened in the last ten minutes. The "suggested" lines are rules, not opinions: the +2 packers line is the Chapter 4.5 arithmetic (pile ÷ hours left); the "3rd unloader" line is the Chapter 4.1 arithmetic. The "spare at hub" column is what tells the dock lead whether holding a trailer is safe tonight.

Screen 2 — Yesterday (refreshed once, 6 AM; counts, percentages, causes)

```
YESTERDAY · Region · Sunday                                                  all three tie-outs: OK

 ┌──────────────────────────────┬──────────┬──────────┬─────────┬──────────────────────────────────────────┐
 │ Measurement                  │ Yesterday│ Plan     │ 7-day   │ Where, and why (click to drill)          │
 ├──────────────────────────────┼──────────┼──────────┼─────────┼──────────────────────────────────────────┤
 │ Arrived on the promised day  │   97.6%  │ ≥97.5    │ ▁▂▃▅▇▇▇ │ Left behind 0.7 at Depot 7 and 12 (Monday)   │
 │ Trucks leaving on time       │   98.1%  │ ≥98      │ ▇▇▇▇▇▇▇ │ 1 of 64: door 14, 20 min late, pack pile     │
 │ Trucks arriving on time      │   95.3%  │ ≥97      │ ▃▂▁▂▃▂▁ │ 3 of 64: two bunching at 1 AM, one slow trip │
 │ Hub window finished          │   98.9%  │ ≥99      │ ▆▆▇▇▇▇▇ │ 1 AM window: 2 trailers rolled               │
 │ Sent to the wrong building   │   0.45%  │ ≤0.30    │ ▁▁▂▅▇▇▇ │ 2 streets on old Depot 3 mapping             │
 │ Left behind at the depot     │   0.7%   │ ≤0.5     │ ▁▁▂▃▅▆▇ │ Depot 7 (210), Depot 12 (190): vans short    │
 │ Trailer fullness, hub trucks │   76%    │ 80       │ ▅▆▇▇▇▇▇ │ 17th trailer at 3 PM and 9 PM since Tuesday  │
 │ Packages per van             │   190    │ 200      │ ▇▇▆▅▃▂▁ │ 11 depots under 195; 2 over 205             │
 │ Packing finished ahead (h)   │   1.3    │ ≥1.5     │ ▇▇▆▅▃▂▁ │ 9 PM truck: 0.9 h; others fine               │
 │ Packages delivered           │ 108,400  │ 104,000  │ ▃▄▅▆▇▇▇ │ Promotion: +4% on re-forecast                │
 └──────────────────────────────┴──────────┴──────────┴─────────┴──────────────────────────────────────────┘
   Click any row → by building → by leave time or route → list of the trailers / vans / packages in the numerator.
```

Screen 3 — This week (Monday 10 AM; dollars, the bridge, the actions)

This is the Chapter 6 page, generated rather than typed: the scorecard table, the five-step bridge from 6.2 computed automatically from the roads' shares and costs, the open-actions list with owners and dates, and a "dollars tied to finance within 0.4%" line at the top. The one-sentence summary is the only thing a person writes.

7.3 What the numbers are made of: scans, trips, routes, and the four tables

Every measurement in this document is a count of packages, trailers, or vans that did or did not do something by a time. So the raw material is the record of what each package, trailer, and van did, and when. That record already exists: every time a package is scanned, a row is written somewhere. The job is to gather those rows into four clean tables.

The scans one package produces (Maria's, from Chapter 2)

#	EVENT	MARIA'S TIME	WHAT IT IS USED FOR	WHICH MEASUREMENT
1	Order placed, promise date set	Mon 2:12 PM	Start of the clock; the promised day	Arrived on the promised day (bottom)
2	Assigned to a warehouse and a truck	Mon 2:12 PM	Which road, which leave time	Share on each road
3	Picked	Mon 2:48 PM	Pick backlog	Today screen, pile
4	Packed	Mon 3:10 PM	Pile for the truck; the last packed time per truck	Packing finished ahead of the truck
5	Labeled	Mon 3:14 PM	The package exists; its destination depot is recorded	Sent to the wrong building (what the label said)
6	Loaded into trailer 14-B	Mon 7:40 PM	Packages aboard each trailer	Trailer fullness
7	Trailer 14-B departed	Mon 9:05 PM	Left on time or not	Trucks leaving on time
8	Trailer 14-B arrived at hub	Tue 1:17 AM	Arrived before the latest useful time or not	Trucks arriving on time
9	Scanned at the hub	Tue 2:10 AM	Inside the window or not	Hub window finished
10	Loaded into the Depot 7 trailer	Tue 3:30 AM	Bagged before the window closed	Hub window finished
11	Trailer departed hub; arrived Depot 7	4:00 AM; 6:25 AM	Second truck on time	Trucks arriving on time
12	Scanned at Depot 7	Tue 7:05 AM	Was this the depot on the label? Did it arrive for today's vans?	Sent to the wrong building; Left behind
13	Loaded on van 17; van departed	Tue 9:04 AM	Packages per van; sent out today or not	Packages per van; Left behind
14	Delivered (or attempted)	Tue 1:53 PM	Was it on or before the promised day?	Arrived on the promised day (top)

Fourteen scans times 100,000 packages is 1.4 million rows a day from one warehouse, tens of millions for a region. That is ordinary for a modern data warehouse, but nobody should compute a dashboard from raw scans every ten minutes. Instead the scans are folded into four tables, once, and every measurement is a simple count on one of them.

The four tables

TABLE	ONE ROW PER	COLUMNS (THE IMPORTANT ONES)	MEASUREMENTS COMPUTED FROM IT
Package journey	Package	Package id; promised date; road; warehouse; truck leave time; the 14 timestamps above; label depot; actual depot; trailer ids; van id; flags computed once: missed truck, late truck, wrong building, left behind, on promised day	Arrived on the promised day and its cause split; wrong building; left behind; share on each road
Trailer trip	Trailer departure	Trailer id; route; scheduled leave; actual leave; packages aboard; positions (2,000); arrival; latest useful arrival for its window; unload start and end; door	Trucks leaving on time; trucks arriving on time; trailer fullness; hub door use
Van route	Van dispatch	Depot; date; wave; van id; packages aboard; stops; departed; returned; failed attempts; cost per route (from the contract rate)	Packages per van; stops per hour; van dollars
Building-day	Building × day	Packages handled; hands-on hours by task; paid hours; dollars (labor from the time system; fixed from finance, by day); planned packages; planned hours	Hub cost per package; depot cost per package; dollars against plan

Two more inputs are not scans. *Invoices*, from carriers and van companies, arrive two to four weeks late; the weekly screen shows an estimated cost (contract rate × measured count) and, when the invoice posts, a separate "true-up" line, so a late bill never looks like a rate change. And *the plan*, loaded per road and per building as planned packages and planned cost per package, which is what makes "against plan" possible.

7.4 One definition card per number

Every tile on every screen points to one card, and the weekly scorecard uses the same card. If two people disagree about a number, the card settles it. If the card changes, the change is dated and the old and new values are shown side by side for four weeks.

Definition card — Trucks arriving on time

FIELD	CONTENT
Question	Of the trucks due in a sorting window, how many arrived early enough to be fully unloaded and sorted before it closed?
Top of the fraction	Trailer trips where arrival time ≤ latest useful arrival. Latest useful arrival = window close − unload time for that trailer's load at the planned crew − 25 minutes.

Bottom of the fraction	Trailer trips scheduled into that window.
Grain	By window, by route; rolled up to day and week.
Source table	Trailer trip.
Owner	Transportation (the road); the hub lead for the crew size that sets the latest useful arrival.
Plan	≥ 97%.
Not to be confused with	Trucks leaving on time (a warehouse measurement). A truck can leave on time and arrive late, or leave late and arrive on time.
Last reconciled	Trip count tied to gate logs exactly, every day; last checked this morning.
Version	2 — crew size made a parameter on 3 November (previously fixed at 2 people).

Definition card — Transportation cost per package

FIELD	CONTENT
Question	What did it cost to move a package from the warehouse dock to the customer's door?
Top of the fraction	Trailer trip costs (route rate × trips) + hub and depot building-day dollars (labor from the time system + fixed from finance) + van route costs + gig driver payments + carrier charges + wrong-building fix costs. Excludes warehouse handling, inbound freight, returns, and refunds, each reported on its own line.
Bottom of the fraction	Packages with a delivery attempt in the week, counted at the delivering depot.
Grain	Week; by region, by road, by package type.
Accrual	Estimated at contract rate × measured count; invoice true-ups shown as their own line when posted.
Allocation	Shared trailers by packages aboard; building fixed cost by packages handled that week.
Owner	This role. Dollars co-owned with transportation finance.
Plan	Per road, loaded from the annual plan; blended plan is computed, never typed.
Last reconciled	Weekly total within 1% of the ledger; last week 0.4%.

7.5 Checking the numbers before anyone sees them

A dashboard that is wrong once is ignored forever. So three checks run automatically before each screen refreshes, and their result is printed at the top of the screen ("all three tie-outs: OK"). If a check fails, the affected tiles still show, greyed, with "not reconciled" and no cause, so nobody acts on a number that might be a data problem.

CHECK	WHAT IS COMPARED	TOLERANCE AND ACTION IF IT FAILS
Packages	Packages in the package journey table for the day against the delivery system's count of attempts for the day	Within 0.5%. Beyond: a feed is missing a depot or a scan type; the data engineer on call is paged; the daily screen marks "Arrived on the promised day" as not reconciled.
Trailers	Trailer trips in the table against the gate logs at each building	Exact. Beyond: a gate scanner is down; the hub's "trucks arriving" tile greys out.
Dollars	Weekly dock-to-door dollars against the ledger for the same week	Within 1%. Beyond: usually a missing invoice feed or a fixed-cost allocation that changed; finance and this role resolve it before the Monday page is published.

The fourth check is human: once a week, one person recomputes one tile by hand from the source table, as Chapter 3 does with a pencil, and compares. Ten minutes a week. It catches the errors that tie out perfectly and are still wrong, like a plan loaded with last year's rates.

7.6 Colors, alerts, and clicking down to the cause

ELEMENT	HOW IT WORKS, AND WHY THAT WAY
Colors	Green within 2% of plan, yellow within 5%, red beyond; for percentages the same in points. The thresholds live on the definition card, not in the chart code, so they can be changed in one place and are the same on all three screens.
Trend marks	Every daily tile shows the last 7 days; every weekly tile the last 6 weeks. Three moves in the same direction trigger a yellow "trend" flag even if every value is green: a problem is cheaper to fix at week three than at week six.
Averages versus tails	Averages hide the trailer that waited an hour for a door (Chapter 4.1). Where a tail matters, the tile shows the count of the worst, not the mean: "trailers past the window: 2," not "average unload time 2.25 hours."
Drill-down	Every tile is a path: region → building → leave time or route → trailer or van → the packages. The last level is a list with the scan history, so "show me the 300 wrong-building packages" is one click, and the two mis-mapped streets in Chapter 6 are found in a minute instead of a meeting.
Alerts, the two-day rule	An alert fires when a daily measurement is red two days in a row, not one, because one day is often noise (one slow truck in 64 is 1.6 points). Exception: the hourly signals on the Today screen fire immediately, because they are about the next few hours.
Alerts, who	Each card names who is told: packing time-left under 1 hour with 3 hours to go → the shift lead; a trailer under 60% at 30 minutes before leave time on a route with 45 minutes of spare → the dock lead, with the hold suggestion; a truck projected to arrive after its latest useful time → the hub window lead and the depot it feeds; a depot whose expected packages exceed 200 × planned vans → the depot manager, at 5 AM, with the gig-booking link.

Actions on the screen	The weekly screen carries the open-actions list from Chapter 6 with owner, date, and the measurement that will close it. A dashboard without an action list is a weather report.

7.7 Who builds what, in what order

WHO	WHAT THEY OWN
This role	The question table (7.1), every definition card, the plan numbers per road and building, the layout of the three screens, the thresholds, the weekly review. The accountable owner of whether the dashboard is trusted.
Data engineers	The pipeline from scan systems to the four tables; the ten-minute refresh of the Today tables; the three tie-out checks; the alerting plumbing.
Data scientists	The expected-arrival time for a truck on the road (from the trip-time record); the package forecast per depot per morning; the "suggested" lines on the Today screen; sensible thresholds from the historical variance.
Transportation finance	The dollar feed, the accrual and true-up rules, the fixed-cost allocation, the ledger tie-out.
Building leads and depot managers	Validation: does the Today screen match what they see from the dock? The cause column's vocabulary. They are the first users and the loudest critics, deliberately.
The reporting tool team	Hosting, access, the drill-down component; not the definitions.

The build, in phases, each with a "done when"

PHASE	WHAT IS BUILT	DONE WHEN	WHY IN THIS ORDER
1	The question table and the definition cards for the twelve measurements; the plan loaded per road	Finance, operations, and the data owner have signed every card. Not a line of pipeline yet.	Everything downstream encodes these; changing them later is the expensive kind of change.
2	The package journey and trailer trip tables, daily refresh	Package and trailer tie-outs pass for ten consecutive days.	These two tables carry nine of the twelve measurements.
3	Screen 3, This week, including the bridge	Three consecutive Mondays where the generated page matches what the team built by hand, and the dollars tie within 1%. Then the hand-built spreadsheet is retired.	The weekly page is where decisions are made; it is also the hardest to fake, so trust is earned here first.
4	Screen 2, Yesterday, with drill-down to the package list	Two regional directors use it in their 8 AM call instead of asking for a report.	Daily causes shorten the weekly meeting.
5	Van route and building-day tables; cost per package by building on the daily screen as an estimate	Estimated weekly dollars within 3% of the invoiced figure for four weeks.	Needed before anyone trusts an estimate on a daily screen.
6	Screen 1, Today: piles and time-left, trailers at the doors, trucks on the road, depot morning view; ten-minute refresh	A dock lead and a hub lead each say they open it instead of calling the other building. Measured: screen opens per shift.	Hourly data is the most work and the most fragile; it goes last so it inherits definitions that already work.
7	Alerts and the suggested-action lines	Alert count per day is under ten for the region and each alert has a named recipient who acted on at least one in the last month.	Alerts added before the screens are trusted get muted.
8	Retire every parallel spreadsheet; measure usage; quarterly definitions review	No number reaches a senior leader from anywhere but the dashboard; usage is on the dashboard itself.	A dashboard that competes with a spreadsheet loses.

7.8 Mistakes that ruin dashboards, and how to avoid each

MISTAKE	WHAT IT LOOKS LIKE	THE GUARD
Too many tiles	Sixty numbers; nobody knows which three matter	Twelve measurements, ranked (Chapter 5.3), on three screens. A new tile must retire an old one or answer a new question on the 7.1 table.
Numbers without owners	A red cell that everyone discusses and nobody fixes	No card without an owner; the owner's name is on the tile.
Denominators that drift	Cost per package "shipped" one week and "delivered" the next; the trend looks noisy and is ignored	The denominator is printed in the tile header; it comes from the card; changing it is a dated version.
Dollars that are half invoiced	A rate that jumps when a late bill posts, read as an operating problem	Estimate at contract rate; true-ups on their own line; the Today screen never shows dollars.
Averages hiding the tail	"Average unload 2.25 hours" when four trailers waited an hour for a door	Show the count past the deadline, and the worst five, next to the average.
Comparing unlike things	This week's cost against a plan built on last year's rates; a pilot depot against the whole region	Reconcile before comparing: same depots, same roads, same week definition; the plan carries its own date.
No path to the cause	A red percentage and a meeting to find out which trailers	Drill-down to the list is built into every tile from phase 4 on.
Alerts nobody reads	Forty alerts a day, muted by week two	Two-day rule, named recipients, under ten a day, and a monthly check that each alert led to an action.
A dashboard nobody opens	Beautiful, accurate, unused	Usage is a tile. If the regional directors are not opening Yesterday by 8:30 AM, the screen is wrong, not the directors.

IF SOMEONE ASKS "HOW WOULD YOU BUILD THE DASHBOARD?" — 199 WORDS

I start with a table of who looks, how often, and what decision they are about to make, and I only build tiles that answer a question on that table. Then I write a definition card for each of the twelve measurements, with the fraction, the owner, the plan, and the source, and get finance and operations to sign them before any pipeline exists. The data is the scans: fourteen per package, folded once a day into four tables, one row per package, per trailer trip, per van route, per building-day. Three tie-outs run before every refresh and print at the top of the screen. There are three screens: today, which is counts and clocks for the people on the dock; yesterday, which is causes for the directors; and this week, which is the scorecard and the mix-versus-rate bridge generated, not typed. Every tile drills to the list of packages behind it. I build the weekly screen first because that is where decisions are made and trust is earned, the hourly screen last because it is the most fragile, and alerts after that, with a two-day rule so they are not muted. And I measure whether people open it.

Chapter 8 — Transportation planning: what is in a trailer, what is in a mile, and how to buy service with trucks

Everything so far costed a truck as "$640 a trip ÷ 1,590 packages." A transportation planner is asked the questions underneath that line: what exactly fills the trailer and what stops it filling further, how full to plan it, why $3.20 a mile and not $2.50, how many trucks a lane needs and when it deserves its own, how many tractors, trailers and drivers that takes, and how you buy a faster promise with trucks. This chapter answers each of those the same way: say what the thing is, say why it matters, do the arithmetic in the open, read what the arithmetic says, name the lever, and end with how to say it in the room. It closes with the scorecard a transportation team runs, the optimization model in the companion workbook, and one machine-learning use case: a camera that measures the air inside a gaylord.

8.1 The transportation question in one sentence

Spend the fewest dollars on trucks such that every package makes its deadline. That is the whole job in one line, and it is worth taking the line apart.

Why "dollars on trucks" and not "dollars on packages." The network does not buy transportation by the package. It buys a truck, for a trip, at a price per mile, and the truck costs the same whether it carries 1 package or 2,000. The package only gets a cost when you divide: $640 for the trip ÷ 1,600 aboard = $0.40. So every cent of transportation cost per package comes from three numbers multiplied together and one number divided: trucks × miles × cost per mile, ÷ packages. That gives exactly three levers, and every transportation decision ever made is one of them.

LEVER	WHAT IT MEANS IN PRACTICE	WHERE THIS PACKET ALREADY USED IT
Fewer miles	Shorter paths: the right hub in the right place; a direct truck where going through the hub is a detour; moving a departure so the truck does not run half the route twice	Chapter 4.6: through the hub is 200 + 120 = 320 miles; direct to Depot 3 is 90
Fuller trucks	More packages sharing each truck's fixed cost: later door-close, fewer departures, fuller gaylords, packages that take less room	Chapter 4.2: 1,600 aboard is $0.40 a package; 800 is $0.80
Cheaper miles	Paying less for the same mile: our own fleet where volume is steady, contract carriers where it is not, bidding lanes once a year, filling the trailer on the way back	Not yet; Section 8.4

Why service is the "such that" and not the goal. Nobody is paid to make trucks fast. They are paid to make trucks cheap while every package still arrives when the website said. In the planner's language the deadline is a *constraint*: a line the plan is not allowed to cross, not a thing to maximize. That distinction decides how every question in this chapter is answered. "Should we add a second departure on this lane?" is never "would it be faster?" (it always would). It is "does the promise need it, and if so what is the cheapest way to buy it?" The deadlines themselves come from Chapter 2.2: the truck has to leave the warehouse at 9 PM to make the hub's 4 AM window, which exists to make the depot's 6:30 AM deadline, which exists to make the vans. Transportation planning is the discipline of meeting that chain of clocks with the fewest truck-dollars.

8.2 What is in a trailer: positions, space, weight, and which one runs out first

Why this comes first. A truck costs $640 whether it carries 1 package or 2,000, so the cost per package on that leg is simply $640 ÷ packages aboard. Everything a planner does about linehaul cost comes down to one question: how many packages can we put on that truck? The honest answer is "it depends on what runs out first." Three things can run out: places to put a container, room, or weight. Which one runs out first is what a planner means by "which constraint binds." The same 53-foot trailer holds very different counts depending on that, and quoting a cost per package without saying which one filled the trailer is how two planners end up arguing about different trucks.

Positions run out first when the freight is in containers

In this network the packages do not ride loose. At the warehouse the sorting machine drops each package into a gaylord: a big cardboard box on a pallet, about 48 inches by 40 by 48, holding packages that all go to one destination. The gaylord is closed when the truck is due to leave, rolled onto the trailer with a pallet jack, and rolled off at the far end with a forklift in minutes. The sort done in the warehouse survives the ride, and the hub does not have to touch each package to know where it goes. That convenience is why the gaylord exists. Its price is air.

Going out: packages → gaylords → trailer (what this packet's trucks carry)

STEP	QUESTION	ARITHMETIC	ANSWER
1	How big is an average package?	A shoebox is about half a cubic foot; a kettle box is one. Average across the mix, with the padding and the air inside the box	**0.6 cubic feet, 2.5 pounds**
2	How big is a gaylord?	48 × 40 × 48 inches = 92,160 cubic inches; ÷ 1,728 cubic inches per cubic foot	**53.3 cubic feet**
3	How many packages fit in a gaylord?	Packages are irregular, they settle unevenly, and the box is closed when the truck is due, not when it is full. About 45% of the space ends up used. 53.3 × 0.45 ÷ 0.6	**40 packages**
4	How many gaylords in a trailer?	A 53-foot trailer has 26 floor spots for a 48 × 40 footprint (13 down each side); a gaylord is low enough to stack two high	**52 gaylords**
5	How many packages in a trailer, by positions?	52 × 40	**2,080, which this packet rounds to 2,000**
6	How much of the trailer's room do those packages use?	2,080 × 0.6 = 1,248 cubic feet of packages, in a trailer with about 3,000 cubic feet of usable room	**42%. The other 58% is air: inside each gaylord, between gaylords, and above the second stack**

Read step 6 again, because it is the point. The trailer is "full" at 2,080 packages with more than half its room empty. What stopped you was not running out of room; it was running out of places to put a gaylord. That is positions binding. It also tells you the lever: when positions bind, the way to get more packages on the truck is to get more packages into each gaylord (close it later, sort more densely, use a taller container), not to buy a bigger trailer.

Room runs out first when the freight is loaded loose

The alternative is to skip the gaylords and stack packages loose, floor to ceiling, wall to wall, nose to tail. Planners call this a fluid load or a floor load. Now there is nothing to run out of except the room itself.

The same trailer, fluid-loaded

STEP	QUESTION	ARITHMETIC	ANSWER
1	How many packages if the room is the only limit?	Boxes never stack perfectly, so allow 85% packing. 3,000 × 0.85 ÷ 0.6	**4,250 packages, about twice the gaylord count**
2	What does that do to cost per package?	$640 ÷ 4,250 against $640 ÷ 2,080	**$0.15 against $0.31 at 100% of positions. Half the cost per package on the same truck**
3	What does it cost at the other end?	Loose packages are unloaded by hand onto a belt at about 1,000 an hour with two people, against the 2 hours Chapter 2 allowed for a gaylord trailer	**About 4 hours at a door instead of 2. Chapter 4.1 priced what two extra hours at a door do to the clock and the doors**
4	What else is lost?	No separation inside the trailer: everything aboard goes to one place, and that place must sort all of it. Bottom layers carry the weight of top layers	**Usable only on thick lanes to one destination that will sort everything anyway; more damage; the fill depends on the loaders that day**

So fluid loading is not "better," it is a trade: half the linehaul cost per package, bought with two extra hours of unload labor and door time, more damage, and the loss of the sort. It pays on a lane like a big warehouse to a big hub, where the hub re-sorts everything anyway and the volume fills trucks either way. It does not pay on a trailer carrying gaylords for ten different depots, because nothing can be loose when the contents have ten destinations. Real networks run both, lane by lane, and a planner is expected to know which lanes are which.

Weight runs out first when the freight is dense

A loaded tractor-trailer may weigh 80,000 pounds on the road. The tractor and the empty trailer are about 36,000 of that, so the payload, the weight of what is inside, may be about 44,000 pounds.

Checking the weight limit

STEP	QUESTION	ARITHMETIC	ANSWER
1	How many packages by weight?	44,000 payload, minus 52 empty gaylords at 30 pounds = 1,560; (44,000 − 1,560) ÷ 2.5	**16,976 packages**
2	What does the full gaylord trailer weigh?	2,080 × 2.5 + 1,560	**6,760 pounds: 15% of the payload. Weight does not bind; positions do, by a factor of eight**
3	And the fluid load?	4,250 × 2.5	**10,625 pounds: a quarter of the payload. Still room, not weight**
4	How heavy would a package have to be for weight to bind at 2,080 positions?	(44,000 − 1,560) ÷ 2,080	**20 pounds. Appliances, pet food, bottled water, cat litter, paper, tile. Then you plan the trailer by pounds, not positions**
5	The rule of thumb	Payload ÷ usable room = 44,000 ÷ 3,000	**About 15 pounds per cubic foot. Freight denser than that "weighs out"; lighter "cubes out." A parcel is 2.5 ÷ 0.6 = 4 pounds per cubic foot, less than a third of the line**

That is what "small parcels almost never hit weight" means. It is not a coincidence of this example; it is the physics of e-commerce boxes, which are mostly air and padding around a light product. The one time a parcel planner does think about weight is the heavy-item mix: a trailer of water and dog food from a warehouse that specializes in bulky goods weighs out at step 4's twenty pounds, and that lane is planned in pounds.

The same question for stock coming into the warehouse

When an interviewer asks "how many units per pallet," they mean the inbound side: product arriving from suppliers in cases on pallets, which is how it gets onto the warehouse shelves in Chapter 2. The hierarchy is units → cases → pallets → trailer, and the arithmetic is the same shape.

Coming in: units → cases → pallets → trailer

STEP	QUESTION	ARITHMETIC	ANSWER
1	Units per case?	The supplier packs 12 kettles in a case	**12 units**
2	Cases per pallet?	A 48 × 40 inch pallet, stacked to 48 inches high so it can ride double-stacked: 12 cases a layer × 5 layers	**60 cases, so 60 × 12 = 720 units a pallet**
3	Pallets per trailer?	26 single-stacked; 52 if the pallets are stable enough to double-stack	**26 or 52 pallets; 18,720 or 37,440 units**
4	Does it weigh out?	A case of kettles is 10 pounds; a pallet 600 pounds plus 50 for the wood; 52 × 650	**33,800 pounds, under 44,000: cubes out at 52**
5	And a heavy item?	Cases of bottled water at 40 pounds: a pallet is 60 × 40 + 50 = 2,450 pounds; 44,000 ÷ 2,450	**17 pallets fit by weight though 52 fit by space: weighs out at 17. A planner who books 26 positions for water has booked 9 empty ones and paid for them**

What binds decides the lever

The reason to insist on naming the binding limit is that the fix for a light trailer is different for each one, and a planner who pulls the wrong lever spends money for nothing.

WHAT RUNS OUT FIRST	WHAT THE PLANNER CAN DO ABOUT IT	WHAT DOES NOT HELP
Positions	Fill the gaylords fuller (close them later, sort more densely), use taller containers, double-stack where not already, or switch a thick single-destination lane to fluid load	A lighter box. The gaylord count is the same whether the packages inside weigh 1 pound or 5
Room	Make the packages smaller: right-sized boxes, bags instead of boxes, more items per package. These are packaging and ordering levers, not dock levers	Anything on the dock. The loaders cannot make a box smaller
Weight	Route heavy goods on their own lanes planned in pounds, mix heavy and light freight on one trailer, or accept fewer units a truck	Anything about containers or stacking. The scale does not care how the freight is arranged

IF SOMEONE ASKS "HOW MANY PACKAGES FIT ON A TRUCK?" — 129 WORDS

It depends on what runs out first, so I say which. For small parcels in gaylords it is positions: twenty-six floor spots, two high, forty packages a gaylord, about two thousand packages, and they use about forty percent of the room and fifteen percent of the weight. Fluid-loaded, room binds and the same truck takes about four thousand, at the price of a four-hour hand unload and no sort inside the trailer. Weight only binds above about fifteen pounds a cubic foot, which parcels never reach. On the inbound side it is twelve units a case, sixty cases a pallet, twenty-six or fifty-two pallets, and water weighs out at seventeen. The limit tells you the lever: positions means fuller gaylords, room means smaller boxes, weight means a different lane.

8.3 How full to plan the trailer: a choice about days, not packages

What the planned fill is. The plan says each trailer on the warehouse-to-hub lane is designed to carry 80% of its positions: 0.80 × 2,000 = 1,600 packages. That one number drives the whole lane plan. It sets the trailer count (11,200 packages for a departure ÷ 1,600 = 7 trailers), the cost per package ($640 ÷ 1,600 = $0.40), the doors and loaders the dock needs, and the number the scorecard measures fill against every week. Chapter 4.2 gave the short reason it is 80% and not 100%. Here is the full reason, because it is the single most-asked transportation question and the answer is a piece of arithmetic, not a habit.

Why it is a choice about days. The real volume for that truck is not 1,600 every day. Mondays are heavier and Saturdays lighter, and the plan already allows for that by day of week. What is left after that is the swing nobody can plan: a promotion, the weather, a big customer's order. Measured over the last year, that swing has a standard deviation of about 10% of the day's plan. In plain words: about two days in three land within 10% of the plan, about nineteen in twenty land within 20%, and one in twenty is further out than that. The truck's roof is fixed at 2,000. So when you choose the planned fill you are really choosing how many days a year the volume fits in the truck. The statistician's word for "the fraction of days below a line" is a quantile, and that is why the detailed packet calls the planned fill a quantile choice.

How often the truck overflows at three planned fills (swing = 10% of the plan; roof = 2,000)

PLAN	AVERAGE	TYPICAL SWING	ROOM TO THE ROOF	SWINGS OF ROOM	DAYS THE VOLUME DOES NOT FIT	PACKAGES THAT DO NOT FIT, AVERAGED OVER ALL DAYS
80%	1,600	160	400	400 ÷ 160 = 2.5	**0.6%: one trailer-day in 160**	0.3 a trailer-day
85%	1,700	170	300	300 ÷ 170 = 1.8	**3.9%: one in 26**	2.7 a trailer-day (about 70 on the days it happens)
90%	1,800	180	200	200 ÷ 180 = 1.1	**13.3%: one in 7.5**	12.1 a trailer-day (about 90 on the days it happens)

How to read "swings of room." Take the gap between the plan and the roof and ask how many typical swings fit in it. At 80% the roof is 400 packages away and a typical swing is 160, so a day has to be two and a half swings above average to overflow. On the normal bell curve a day that far out happens 0.6% of the time. At 90% the roof is only 200 away and a swing is 180, so a day barely over one swing above average overflows, and that happens 13% of the time. The last column counts zero on the days that fit, so "12 packages a trailer-day" at 90% means that on the one day in seven or eight when it overflows, about 90 packages are left on the dock.

The same curve, read as a 95% band and a 99% band. A 95% band is the range of package counts that holds 95 days out of 100. It is drawn the same way every time: start at the average, go 1.96 typical swings up, and go 1.96 typical swings down. A 99% band uses 2.576 swings each way, which is 2.58 to two decimals; the tables keep 2.576 so the package count rounds cleanly. These two numbers, 1.96 and 2.58, are properties of the bell curve and do not depend on the truck. Because the band has two ends, the days that fall outside it split half and half. So 2.5 days in 100 land above the top of a 95% band, and 0.5 days in 100 land above the top of a 99% band.

Worked for the 80% plan: average 1,600, swing 160, roof 2,000

STEP	ARITHMETIC	RESULT
1. Half-width of the 95% band	1.96 × 160	**314 packages**
2. The 95% band	1,600 − 314 to 1,600 + 314	**1,286 to 1,914. 95 days in 100 land in here**
3. Days above the top	(100 − 95) ÷ 2	**2.5 days in 100 are above 1,914**
4. Half-width of the 99% band	2.576 × 160	**412 packages**
5. The 99% band	1,600 − 412 to 1,600 + 412	**1,188 to 2,012. 99 days in 100 land in here**
6. Days above that top	(100 − 99) ÷ 2	**0.5 days in 100 are above 2,012**

7. Where the roof is	2,000 − 1,914 = 86 above the 95% top; 2,012 − 2,000 = 12 below the 99% top	**The roof sits between the two tops, so the overflow share sits between 2.5% and 0.5%. The first table's 0.6% is that number**

The 95% band and the 99% band at all three plans (roof = 2,000 throughout)

PLAN	AVERAGE	95% OF DAYS FALL BETWEEN	99% OF DAYS FALL BETWEEN	WHERE THE ROOF SITS
80%	1,600	1.96 × 160 = 314, so 1,286 to 1,914	2.576 × 160 = 412, so 1,188 to 2,012	**86 above the 95% top, 12 below the 99% top. Overflow between 0.5% and 2.5%: it is 0.6%**
85%	1,700	1.96 × 170 = 333, so 1,367 to 2,033	2.576 × 170 = 438, so 1,262 to 2,138	**33 below the 95% top. Overflow more often than 2.5%: it is 3.9%**
90%	1,800	1.96 × 180 = 353, so 1,447 to 2,153	2.576 × 180 = 464, so 1,336 to 2,264	**153 below the 95% top. Overflow well above 2.5%: it is 13.3%**

The one mix-up to avoid. "95% of days fit under the roof" and "the roof is the top of the 95% band" are different lines, because the band spends half of its leftover days on the light side. The top of the 95% band has 2.5% of days above it, so 97.5% of days are under it. The line with exactly 95% of days under it, and 5% above it, is only 1.645 swings above the average. The line with exactly 99% of days under it is 2.326 swings above the average. Set the roof equal to that line and solve for the plan:

The plan that makes the roof cover 95% of days, and the plan that covers 99%

SHARE OF DAYS UNDER THE ROOF	SWINGS ABOVE THE PLAN	ARITHMETIC	THE PLAN
95% (5% overflow)	1.645	2,000 ÷ (1 + 1.645 × 0.10) = 2,000 ÷ 1.1645 = 1,717	**1,717 packages: 85.9% of the roof**
97.5% (the top of the 95% band)	1.960	2,000 ÷ (1 + 1.960 × 0.10) = 2,000 ÷ 1.1960 = 1,672	**1,672 packages: 83.6% of the roof**
99% (1% overflow)	2.326	2,000 ÷ (1 + 2.326 × 0.10) = 2,000 ÷ 1.2326 = 1,622.6	**1,623 packages: 81.2% of the roof**
99.5% (the top of the 99% band)	2.576	2,000 ÷ (1 + 2.576 × 0.10) = 2,000 ÷ 1.2576 = 1,590	**1,590 packages: 79.5% of the roof**

Read the last row against the first table. An 80% plan puts the roof almost exactly at the top of the 99% band, which is why that row overflows about 0.5 days in 100 (the measured figure is 0.6). Planning 1,717 a truck, 85.9% of the roof, is what it takes for 95 days in 100 to fit. The 90% plan covers 87 days in 100.

The two costs being traded

Plan low, and every day pays for air. At 80% the truck carries 1,600 for $640. If it carried 1,800, the same 1,600 packages would need only 1,600 ÷ 1,800 = 0.89 of a trailer, so the network would buy 11% fewer trucks. The saving is (1 − 1,600 ÷ 1,800) × $640 = $71 a day for every 1,600 packages, which is the same as 4.4 cents a package ($0.400 against $0.356). The point is that this saving is given up every single day, including the 99.4% of days when the extra room was never needed. That is what "paying for air" means: the empty fifth of the trailer is bought on purpose, as insurance.

Plan high, and some days overflow. On the days the volume beats the roof, the packages that do not fit go one of two ways. Either the dock calls for an extra trailer, which costs $640 each time it happens, or the packages roll to the next departure, which costs about $2 a package in re-handling and re-planning and, if the next departure is too late for the promise, a missed delivery day. The expected cost of each is the chance of it happening × what it costs when it does.

Saving against overflow cost, per 1,600 packages a day, compared with the 80% plan

PLAN	TRUCK DOLLARS SAVED	IF THE OVERFLOW MEANS AN EXTRA TRAILER	IF THE OVERFLOW ROLLS TO THE NEXT TRUCK
80%	$0	0.6% × $640 = **$4.0**	0.3 × $2 = **$0.6**
85%	(1 − 1,600 ÷ 1,700) × $640 = **$37.6**	3.9% × $640 = **$24.8**	2.7 × $2 = **$5.3**
90%	(1 − 1,600 ÷ 1,800) × $640 = **$71.1**	13.3% × $640 = **$85.3**. Costs more than it saves	12.1 × $2 = **$24.3**. Saves $47, but 12 packages a day are at risk of being late

Reading the table. Go across the rows. If an overflow has to be cleared by a whole extra truck, 85% is the best of the three (saves $37.6, costs $24.8) and 90% loses money (saves $71.1, costs $85.3): the occasional extra truck eats the daily saving. If the overflow can roll to a later departure, 90% wins on dollars ($71.1 against $24.3), but you are accepting about 12 packages a trailer-day that will ride a later truck, and whether that is acceptable has nothing to do with money. It depends on whether the later truck still makes the promise.

The decision rule. Plan the fill where truck dollars saved minus expected overflow cost is largest, subject to the roll not breaking the promise. That turns on one fact about the lane, and it is the fact Chapter 2.2's clock chain gives you: is there a later departure that still arrives before the deadline?

- **Yes, the lane has slack.** Plan toward 90% and let the occasional overflow roll. The roll is cheap and nobody downstream notices. Most warehouse-to-hub lanes with several departures a night are like this for the early departures.
- **No, this is the last truck that makes the promise.** Plan near 80% and dispatch on time. The air is the price of the promise, and it is a cheap price: 4 cents a package to protect the delivery day.

The same arithmetic in one line. This is the same decision as how many newspapers a stand should stock when it does not know the day's sales. The share of days you should allow to overflow equals the cost of one empty slot divided by the cost of one empty slot plus the cost of one overflowed package. An empty slot on this lane is worth about $640 ÷ 1,800 = $0.36. If an overflowed package costs $2 to roll, the answer is $0.36 ÷ ($0.36 + $2.00) = 15% of days, which is the 90% plan almost exactly. If an overflowed package has to be carried by an extra truck, its cost is many times higher and the answer slides back toward the 80% plan. A planner who can say that in one breath has explained the fill target better than any policy document.

Why the scorecard carries both fill and the roll. A fill number on its own lies. A lane at 90% fill looks efficient, but if it is the last departure that makes the promise it is rolling packages onto late trucks one day in seven. A lane at 78% looks wasteful, but if it is a last-departure lane it is doing exactly its job. So the fill

target is set lane by lane from the slack in the clock, never as one number for the network, and the scorecard prints fill and "packages that missed their planned departure" side by side.

IF SOMEONE ASKS "WHY DO YOU PLAN 80% AND NOT 100%?" — 153 WORDS

Because the plan fill is a choice about days, not packages. Daily volume swings about ten percent around the plan and the roof is fixed at two thousand. At eighty percent the roof is two and a half swings away, so the truck overflows one day in a hundred and sixty. At ninety it is one swing away and overflows one day in seven. Going from eighty to ninety saves seventy dollars a day per sixteen hundred packages, about four cents a package. If an overflow means an extra truck, that costs eighty-five dollars a day in expectation, so ninety loses. If the overflow can roll to a later truck that still makes the promise, it costs twenty-four, so ninety wins. So I set the fill per lane from the slack in the clock: plan high and roll where there is a later truck, plan near eighty and dispatch where there is not.

8.4 What is in the $3.20 a mile, and how cost per mile becomes cost per package

Why take the rate apart. Every truck row in this packet used $3.20 a mile as a given. A planner cannot treat it as given, because the rate is the one lever of the three that is bought rather than designed, and buying it well means knowing what is inside it. There are two surprises inside: the biggest single piece is the driver, not the fuel, and a quarter of the price is for miles the package never rides.

The $3.20 a loaded mile, taken apart (illustrative; a truckload rate in a normal year is $2.00–$3.00 per loaded mile before the empty return)

PIECE	WHAT IT PAYS FOR, AND THE ARITHMETIC	$ PER MILE DRIVEN
Driver	Wages and benefits of about $30 an hour. At 50 miles an hour on the road that is $30 ÷ 50 = $0.60 a mile. Add the paid hours waiting at docks and in yards, which produce no miles but must be paid for by the miles that are driven	$0.75
Fuel	A loaded tractor gets about 7 miles to the gallon. At $4 a gallon that is $4 ÷ 7 = $0.57. This is the piece that moves with the market, which is why contract rates carry a separate fuel surcharge that rises and falls with the price of diesel	$0.57
Tractor	Lease or depreciation of about $2,500 a month, spread over about 9,000 miles a month: $2,500 ÷ 9,000 = $0.28. A tractor that runs fewer miles costs more per mile, which is the whole case for keeping trucks busy	$0.28
Trailer	Lease and tires on a trailer pool that is bigger than the tractor pool, because trailers wait at docks while tractors keep moving (Section 8.6: about 60 trailers for 35 tractors)	$0.12
Maintenance and tires	Scheduled service, repairs, a set of tires every 100,000 miles or so	$0.20
Insurance, permits, tolls	Liability and cargo insurance, road taxes, toll roads on the route	$0.18
Carrier overhead and margin	If a carrier runs the truck for us: dispatchers, safety, back office, and profit. If we run our own fleet this line is our own overhead instead	$0.30
Cost per mile driven	**What it costs to move the truck one mile, loaded or empty**	**$2.40**
The empty return	Our lanes are out-and-back. If every trailer came home empty, half of all miles would be empty. A planner who finds a return load for half the trips (inbound freight to the warehouse, empty gaylords and totes going back) gets the empty share down to a quarter of all miles. The empty miles still have to be paid for, and the only thing that can pay for them is the loaded miles: $2.40 ÷ (1 − 0.25) = $3.20	+ $0.80
Cost per loaded mile, what the package pays	**The rate every truck row in this packet used**	**$3.20**

The empty return, slowly. A truck that drives 200 miles to the hub and 200 miles back has driven 400 miles. If it carried packages only on the way out, the 200 loaded miles have to cover the cost of all 400: $2.40 × 400 ÷ 200 = $4.80 a loaded mile. If the planner finds something to carry back on half the trips, then out of every 400 miles 300 are loaded and 100 empty, the empty share is a quarter, and the loaded mile pays $2.40 × 400 ÷ 300 = $3.20. If every trip came back loaded, the loaded mile would be $2.40. So the same truck, driver and fuel cost $2.40, $3.20, or $4.80 a loaded mile depending on one thing: how often it comes back full. That is why "cost per mile" is never a complete number on its own.

Three ways to buy the mile. A *dedicated fleet* (our tractors and our drivers, or a carrier's assigned to us full time) costs about $2.40–$2.60 a mile all-in when it is kept busy: there is no middleman's margin, and the planner controls both directions so return loads are easier to arrange. But the cost is paid whether the truck runs or not, so it only pays on lanes with steady daily volume, which is exactly what the warehouse-to-hub lanes are. A *contract carrier* is paid per load at a lane rate bid once a year, about $3.00–$3.40 a loaded mile; it absorbs the day-to-day variation for us, and the risk is that in a tight market the carrier declines a load it agreed to carry (the share it accepts is the "tender acceptance" line on the scorecard), and the load goes to the *spot market* at $4.00–$6.00 a mile, whatever a truck costs that day. A network usually runs dedicated on its thick, steady lanes, contract carriers on the rest, and spot as the safety valve it hopes not to use.

From cost per mile to cost per package: the same 200-mile lane, cut six ways

CHANGE	ARITHMETIC (WAREHOUSE → HUB, 200 MILES)	COST PER PACKAGE
As is	$3.20 × 200 ÷ 1,600	$0.400
Fuller: 1,800 aboard (Section 8.3's trade)	$3.20 × 200 ÷ 1,800	$0.356, saves 4.4 cents
Cheaper mile: dedicated fleet at $2.60	$2.60 × 200 ÷ 1,600	$0.325, saves 7.5 cents
Fewer miles: a hub 20 miles closer to the warehouse	$3.20 × 180 ÷ 1,600	**$0.360, saves 4 cents, but every hub → depot leg changes too; this is the network design question**

Better return loads: empty share from a quarter to a tenth	$2.40 ÷ (1 − 0.10) = $2.67 a loaded mile; $2.67 × 200 ÷ 1,600	$0.333, saves 6.7 cents
No return loads at all: empty share half	$2.40 ÷ (1 − 0.50) = $4.80 a loaded mile; $4.80 × 200 ÷ 1,600	$0.600, costs 20 cents more

Reading the table. The three levers of Section 8.1 are all here, and they are worth about the same: 4 to 7 cents a package each on a 40-cent leg. The last two rows are the trap. Cost per loaded mile already contains an assumption about empty miles, so two planners quoting "$3.20" may mean different things: the same truck is $2.67 with good return loads and $4.80 with none. The empty return is worth as much as the fill. That is why the scorecard in Section 8.8 carries *empty miles share* as its own line and the rate is always quoted together with it.

IF SOMEONE ASKS "WHAT IS IN YOUR COST PER MILE?" — 128 WORDS

About two dollars forty to move the truck a mile: seventy-five cents of driver, which is the biggest piece, fifty-seven of fuel, forty of tractor and trailer, forty of maintenance, insurance and tolls, and thirty of carrier overhead if a carrier runs it. Then the empty return. Our lanes are out-and-back, and with return loads on half the trips a quarter of the miles are empty, so the loaded miles pay two forty divided by three quarters, which is three twenty. I always quote the rate with the empty share, because the same truck is two sixty-seven with good return loads and four eighty with none. I buy the mile three ways: dedicated on the thick steady lanes, contract carriers on the rest, spot as the safety valve.

8.5 Lane planning: how many trucks a day, and when a direct truck beats the hub

What a lane is. A lane is one origin and one destination with a schedule: Warehouse A to Hub West, leaving at 9 PM, every day. The network is a list of lanes, and planning it is three questions asked of each one: how many packages a day, how many trucks a day that takes, and whether those trucks will be full enough to be worth running, or whether the packages should instead ride a shared lane through a hub. The last question is where network design and transportation planning meet, and the arithmetic is simpler than it sounds.

The idea in one sentence. A direct truck costs the same every day whether it is full or a fifth full; a hub path costs a fixed amount per package. So a direct truck is worth running when the daily volume is large enough that its fixed daily cost, spread over the packages, is less than what the hub path would have charged each of them. That break-even volume is simply the truck's daily cost ÷ the hub path's cost per package. Lanes above it are "thick" and should run direct; lanes below it are "thin" and should ride the hub.

A thick lane and a thin lane, from the companion workbook's region (4,000 packages a day across 12 zones)

STEP	QUESTION	ARITHMETIC	ANSWER
1	Downtown gets 4,400 packages a week from Warehouse A, 145 miles away. Trucks a day for a direct lane?	4,400 ÷ 7 = 629 a day; ÷ 2,080 a trailer = 0.30 of a trailer. But next-day service needs a departure every day, and you cannot run three tenths of a truck	**1 truck a day, 30% full**
2	What does that lane cost?	Each truck-day is 145 miles × $3.20 = $464. A week is 7 × $464 = $3,248; ÷ 4,400 packages	**$0.74 a package**
3	What does the hub path cost the same packages?	Warehouse A → Hub West, 65 miles on a shared trailer at 80% fill: $3.20 ÷ 1,664 = $0.0019 a package-mile, × 65 = $0.125. Hub handling $0.16. Hub West → Downtown, 80 miles = $0.154. Hub West's $4,000 a week of fixed cost spread over its 16,000 packages = $0.25	**$0.69 a package**
4	Break-even volume for a direct lane	Truck cost per day ÷ hub-path cost per package = $464 ÷ $0.69	**672 packages a day. Downtown's 629 is just under, so the model keeps Downtown on the hub when cost is all that matters, and moves it to a direct truck the moment same-day service is required (Section 8.9)**
5	Farmlands gets 1,600 a week from Warehouse A, 75 miles. Direct?	229 a day, 11% of a trailer. 7 × 75 × $3.20 = $1,680 a week ÷ 1,600	**$1.05 a package direct, against $0.125 + 0.16 + 0.19 + 0.25 = $0.73 through the hub. The hub wins by a third. Thin lanes are what hubs are for**
6	The same question in the big network of Chapter 3 (100,000 a day)	A zone taking 8,000 a day from one warehouse fills 8,000 ÷ 2,080 = 3.8 trailers: 4 trucks a day at 96%	**Direct wins outright. Whether a hub earns its keep is a question about lane thickness, and the answer flips with scale. This is why big warehouses run direct trucks to big delivery stations and hubs carry everything else**

Reading the table. Step 4 is the number to remember: on a 145-mile lane, a direct truck pays for itself at about 670 packages a day, which is a third of a trailer. It is not "a full trailer," as people often assume, because the alternative is not free: every package on the hub path pays 69 cents, and 670 of them pay $464, which is what the truck costs. The break-even moves with distance (a 75-mile lane breaks even at $240 ÷ $0.73 = 330 a day) and with the hub's own cost (a cheaper hub raises the break-even and keeps more lanes consolidated). Step 6 is the second thing to remember: the same zone that rides the hub in a 4,000-a-day region runs four direct trucks in a 100,000-a-day one. Hubs are not good or bad; they are the right answer below a volume and the wrong answer above it.

The schedule is part of the lane. One truck a day leaving at 9 PM serves next-day. The same lane serving same-day needs a second departure late in the morning, so the lane's cost doubles while its volume does not, and the fill of each truck halves. The departure times come from the deadline chain in Chapter 2.2, working backward from the van wave each one feeds. The number of departures comes from the volume. The fill then falls out as volume ÷ (departures × packages per trailer), and it is the number the planner watches, because a departure added for a Tuesday surge and never removed shows up as a lane whose fill has quietly dropped from 80% to 55% (Chapter 6's "17th trailer added and never removed").

IF SOMEONE ASKS "WHEN DOES A LANE DESERVE ITS OWN TRUCK?" — 131 WORDS

When its daily volume beats the truck's daily cost divided by what the hub path charges a package. On a hundred-and-forty-five-mile lane the truck is four hundred sixty-four dollars a day and the hub path is sixty-nine cents a package, so the break-even is about six hundred seventy packages a day, a third of a trailer, not a full one. Downtown at six hundred twenty-nine is just under and rides the hub; Farmlands at two hundred thirty would pay a dollar five direct against seventy-three cents through the hub. Scale flips the answer: the same zone at eight thousand a day fills four trucks and goes direct. And the schedule is part of the lane: a same-day promise means a second departure, which doubles the lane cost and halves the fill.

8.6 Tractors, trailers and drivers: sizing the fleet for one lane

Why fleet size follows from the clock. Once a lane has its trips a day, the question becomes what it takes to run them: how many tractors, how many trailers, how many drivers. None of these is a guess. Each one is trips a day × how long each trip ties the asset up, ÷ 24 hours, plus an allowance for the asset being unavailable. The three answers are different because the three assets are tied up for different lengths of time: a tractor is tied up for the round trip, a trailer also for the loading and unloading at both ends, and a driver for the round trip but only within the hours the law allows.

The warehouse → hub lane from Chapter 3: 64 trailers a day, 200 miles each way

STEP	QUESTION	ARITHMETIC	ANSWER
1	How long is one round trip for a tractor?	4 hours out + 2 hours while the trailer is unloaded (or 0.5 if the driver drops the full trailer and hooks an empty one) + 4 hours back + 1 hour in yards and fueling	**11 hours; 7.5 with drop-and-hook**
2	Round trips per tractor per day?	24 ÷ 11 = 2.2, less the time for inspections and the fact that departures are at fixed hours	**2 a day**
3	Tractors?	64 trips ÷ 2 a day = 32, plus about 10% so that a tractor in the shop does not cancel a departure	**35 tractors**
4	Trailers?	Trailers in motion or being unloaded at any moment: 64 × 11 ÷ 24 = 29. Trailers sitting at warehouse doors being loaded for the next departure: 16. Empty trailers waiting at the hub to come back: 16	**About 60 trailers, nearly two per tractor. Trailers are cheap to own and expensive to run short of: a departure with no trailer at the door is a missed departure**
5	Drivers? A driver may drive 11 hours and be on duty 14 in a day, then must rest 10	One round trip is 8 hours of driving and 11 on duty, inside both limits: one driver, one round trip, one shift. 64 round trips a day need 64 driver-shifts	**64 driver-shifts a day. With two days off a week, holidays and sickness, the roster is about × 7 ÷ 5 = 90 drivers**
6	What if the lane were 350 miles?	7 hours out and 7 back is 14 hours of driving, over the 11-hour limit	**Either two drivers share the tractor, or two drivers meet at the midpoint and swap trailers, or one driver sleeps at the far end and the tractor does one trip a day instead of two, which doubles the tractor count. Lane length changes fleet cost in steps, not smoothly**

Reading the table. Three things an interviewer listens for. First, that the trailer count is about double the tractor count, and why: a tractor should never wait for a trailer to be loaded, so the trailers do the waiting instead, at both ends. Second, that drop-and-hook (leave the full trailer, take an empty) cuts the tractor cycle from 11 hours to 7.5, which is a third fewer tractors and drivers for the same trips, paid for with the spare trailers of step 4. Third, the step change at step 6: the hours-of-service rule means a lane just over about 275 miles each way (5.5 hours of driving each way) cannot be a one-driver round trip in a day, and its fleet cost roughly doubles. Network designers who put a hub 300 miles from a warehouse to save a building have sometimes bought that doubling without noticing.

The utilization idea. Everything in this table assumes the assets are busy: a tractor doing two round trips a day, a driver on a full shift. A tractor that does one round trip a day costs the same $2,500 a month but spreads it over half the miles, so its line in Section 8.4 goes from $0.28 to $0.56 a mile. This is the whole argument for a dedicated fleet on steady lanes and against it on lanes that run three days a week: the dedicated truck is cheap per mile only when the miles are there every day.

8.7 Service level in transportation terms: how you buy a faster promise with trucks

What service level means to a transportation team. Of the trucks that had a deadline, how many beat it. Chapter 4.1 did the arithmetic for one lane: 90 trips in 100 are under 4 hours 27 minutes; the hub needs the truck by 1:36 AM; 4 in 100 are late. Every way to raise that number costs something, and the planner's skill is knowing the menu and the prices, because the cheapest item is different on every lane.

WAY TO BUY SERVICE	WHAT IT DOES TO THE CLOCK	WHAT IT COSTS
Leave earlier	Adds spare time at the far end: a 9 PM departure that leaves at 8:30 arrives with 30 more minutes against the 4 AM window	Nothing in truck dollars. But the customers' order deadline moves 30 minutes earlier, and some orders that would have made tonight's truck go a day slower (Chapter 4.1). Service on the road is bought with service on the website
Unload faster (three-person crews, drop-and-hook)	Moves the "latest useful arrival" later: if unloading takes 1 hour instead of 2, a truck arriving at 2:36 still makes the window	Labor hours about the same, staffing shape different; drop-and-hook needs the spare trailers of Section 8.6
Add a departure	A package that misses one truck waits hours instead of a day; a late truck's packages catch the next one	One more truck a day: $640 × 360 days = $230,000 a year on a 200-mile lane, and the fill of every departure on the lane falls
Skip the hub (direct truck)	Removes the 3-hour sort window and one leg: Chapter 4.6's 9.5 hours become 2. The only way to reach same-day over any distance	Cheap on thick lanes, expensive on thin ones (Section 8.5): a 30%-full truck every day

Faster mode (two drivers, expedited)	A 700-mile leg in 12 hours instead of 24, because the truck never stops for the driver's rest	Roughly double the cost per mile. Used on a few lanes that feed a promise, never for the base network
Outside carrier for the far tail	Their network, their schedule	$4.50 a package against $2.58. Buys reach to places our network does not go; buys two-day, not speed

How the workbook turns this into numbers. The companion model does not simulate late trucks; it asks a simpler question of every path: can a package that leaves the warehouse dock on this path be at the delivery station in time for the van wave? A path is *same-day eligible* if that takes 4 hours or less (truck leaves late morning, package is on an afternoon van) and *next-day eligible* if 9.5 hours or less (truck leaves at 9 PM, package is at the depot by 6:30 AM). Through a hub the hours are drive + 3-hour sort window + drive, so same-day through a hub is impossible at any realistic distance and next-day needs the two drives to total under 6.5 hours. Direct, hours = drive, so same-day reaches about 175 miles at 50 miles an hour with half an hour in yards. "Buy 30% same-day" then means "put 30% of the packages on direct trucks," which means thin lanes running a daily truck, which has a price the model computes in Section 8.9.

> **IF SOMEONE ASKS "HOW DO YOU RAISE ON-TIME WITHOUT SPENDING MONEY?" — 126 WORDS**
>
> You can, but you pay somewhere else. Leaving half an hour earlier costs no truck dollars and moves the customer cutoff half an hour earlier, so some orders go a day slower. Unloading faster moves the latest useful arrival later and costs a staffing change and spare trailers. After that every item has a price: a second departure is two hundred thirty thousand a year on a two-hundred-mile lane and halves the fill; a direct truck removes the three-hour sort window and is cheap only on thick lanes; team drivers double the cost per mile. So I treat on-time as a constraint, list the menu for that lane, and buy the cheapest item that meets the promise, which is a different item on almost every lane.

8.8 The transportation scorecard

This is the page the transportation team reads on Monday, underneath the network page of Chapter 6. Same rules: every number is a fraction with its bottom printed, it has a plan, an owner, and a cause written beside every red cell. The "our model" column is the companion workbook's baseline region, so every value can be traced to a cell. The rows are in four groups, and the groups are the way to read the page: money, then how the trucks were used, then whether they ran on time, then how the miles were bought.

MEASUREMENT	THE FRACTION	OUR MODEL	PLAN	OWNER	WHEN IT IS RED, LOOK AT
Money					
Transportation cost per package (dock to door)	dock-to-door dollars ÷ packages delivered	$2.35	$2.40	This role	Mix versus rate (Chapter 6.2): did the packages change or did the prices
Cost per loaded truck mile	linehaul dollars ÷ loaded miles	$3.20	$3.20	Procurement	Fuel surcharge; spot loads; the empty-mile share below
Cost per package-mile	linehaul dollars ÷ the sum over all packages of the miles each one rode	$0.0019	$0.0022	Network design	Rises with thin direct lanes and light trailers. The single best summary of middle-mile efficiency, because it is cost per mile ÷ packages per truck in one number
How the trucks were used					
Truck miles a week	sum over lanes of trips × lane miles	2,520	plan by lane	Planning	Departures added for a surge and never removed
Trailer trips a week	count of departures	34	plan	Planning	Each trip over plan is $640–$930
Packages per trailer trip	package-trips ÷ trailer trips	1,664	1,660	Planning	Under 1,500: a departure should come off
Fill, consolidated lanes (by positions)	packages aboard at door-close ÷ packages per trailer	80%	80%	Dock leads	Under 75% three weeks running: lane review. Over 90%: check the roll line below before celebrating (Section 8.3)
Fill, direct lanes	direct packages ÷ (trucks × days × packages per trailer)	—	≥ 40%	Network design	Under 40% unless the lane is buying same-day service
Packages that missed their planned departure (the roll)	packages rolled to a later truck ÷ packages planned	measured	≤ 0.5%	Dock leads	A fill target set too high for the lane's slack (Section 8.3)
Whether they ran on time					
Trucks leaving on time	departures within 15 minutes of schedule ÷ scheduled	measured	≥ 98%	Warehouse dock	Chapter 4.5: packing not finished; trailer not at the door
Trucks arriving before the latest useful time	arrivals before the deadline ÷ trips due	measured	≥ 97%	Road and hub	Chapter 4.1: road time; door bunching at the hub
Dwell at dock	sum of (unload start − arrival) ÷ number of arrivals	measured	≤ 30 min	Hub and depot leads	Door bunching; crews; spread the departures
How the miles were bought					
Empty miles share	empty miles ÷ all miles	25% (assumed)	≤ 25%	Planning	Pair lanes with inbound or returns. Every 5 points of empty share is about 20 cents on the loaded mile
Tender acceptance (contract carriers)	loads accepted at contract rate ÷ loads offered	measured	≥ 95%	Procurement	Under 90% the spot market is pricing you: re-bid the lane or go dedicated
Spot share of loads	loads bought on the spot market ÷ all loads	measured	≤ 3%	Procurement	Rises when tender acceptance falls; each spot load is roughly $1.50 a mile over contract
Same-day share · next-day share · average hours	packages on eligible paths ÷ all; package-weighted hours dock to depot	0% · 100% · 7.0	scenario	This role	Section 8.9: the service the network was designed to give

Damage and loss per 1,000	claims ÷ packages × 1,000	measured	≤ 0.5	Hub and transportation	Gaylord fill rules; load bars; fluid-load stacking; chute speed

How the page rolls up. Cost per loaded mile × miles ÷ packages is the linehaul part of transportation cost per package, so the money group is explained by the other three. Fill and departures are why miles per package move; empty share and tender acceptance are why cost per mile moves; on-time departure and arrival are why the network page's "arrived on the promised day" moves. When Chapter 6's page shows cost up twelve cents, this page says which of the three levers slipped: more miles (a lane added, a hub moved), lighter trucks (fill down, departures up), or dearer miles (spot share up, empty share up).

A worked red cell. Suppose Monday's page shows cost per loaded mile at $3.41 against the $3.20 plan, with tender acceptance at 88% and spot share at 6%. Read it as a chain: the market tightened, contract carriers declined 12% of loads instead of 5%, 6% of loads went to spot at about $4.70, and 6% × ($4.70 − $3.20) = $0.09 of the 21-cent rise is spot. The rest, 12 cents, is the fuel surcharge, which the fuel line confirms. The action is not "negotiate harder"; it is to re-bid the two lanes with the worst acceptance and to ask whether the thickest of them should go dedicated, which Section 8.6 said pays when the volume is steady. That is what an owner column is for: the row tells procurement what to do on Tuesday.

8.9 The optimization model in the companion workbook

The workbook *outbound-network-design-model.xlsx*, in the same folder, holds two models you can run with Excel's Solver. The first is the uploaded root-beer transshipment baseline (plants → distribution centers → customers), rebuilt with formulas and pre-solved: $78,524 a week, $39.26 a barrel, five centers open, the level-of-service rules met (average 23.5 miles against a 25-mile maximum; 86.5% of barrels within 50 miles against an 80% minimum). The second is that same structure turned into an outbound transportation network, and this section is its plain-English description.

What "optimization model" means here. The planner writes down every path a package could take, what each path costs per package and how many hours it takes, and every rule the plan must obey. Then Solver, which is a search procedure built into Excel, finds the mix of paths that costs least while obeying every rule. The planner's job is not to find the answer; it is to write the paths, costs and rules honestly, and then to read the answer and say why it came out that way. Every scenario below is the same model told one different thing.

What the model decides, and what it is told

PIECE	CONTENT
Decides	How many packages a week ride each path (warehouse → hub → zone for 2 warehouses × 5 hubs × 12 zones; warehouse → zone direct; warehouse → outside carrier); how many trucks a day run on each direct lane (whole numbers, because you cannot run a third of a truck); which hubs are open (yes or no). 185 cells for Excel's Solver to change.
Minimizes	Cost per package = (warehouse handling + inbound linehaul + hub fixed cost + hub handling + outbound linehaul + direct-lane trucks + carrier charges + delivery station and van) ÷ packages.
Subject to	Every zone's demand delivered. A hub sorts at most its capacity, and only if open. A warehouse ships at most its capacity. A direct lane carries at most trucks a day × 7 × packages per trailer. Hubs open between a minimum and maximum. Same-day share ≥ target. Next-day share ≥ target. Average hours ≤ target. Direct lanes and the carrier allowed or not.
Is told	Map coordinates of 2 warehouses, 5 candidate hubs, 12 zones (distances are formulas, so a hub can be moved and everything recomputes); $3.20 a mile; the trailer build-up of Section 8.2; 80% planned fill on consolidated lanes (Section 8.3); speeds and yard times; the 4.0 and 9.5-hour eligibility rules of Section 8.7; hub capacities, fixed costs, handling rates; zone demand and van cost by density; the $4.50 carrier.
The one modeling choice to defend	Consolidated hub lanes are costed per package at planned fill, because those trucks carry other network volume outside this 4,000-a-day slice. Direct lanes are dedicated to one warehouse and one zone, so they pay for whole trucks every day. That asymmetry is the economics of Section 8.5, and it is what makes the model choose hubs for thin flows and direct trucks for thick ones or for same-day service. If an interviewer asks "why did the model open a hub," this is the answer.

The twelve scenarios, solved

#	SCENARIO	COST PER PACKAGE	DOCK TO DOOR	HUBS OPEN	SAME-DAY	NEXT-DAY	DIRECT SHARE	WHAT IT SAYS
1	Lowest cost, no service rule	$3.123	$2.351	2 (West, East)	0%	100%	0%	Two hubs, nothing direct
2	Next-day ≥ 90%	$3.123	$2.351	2	0%	100%	0%	Next-day is free at this geography
3	Next-day ≥ 95%, same-day ≥ 15%	$3.160	$2.396	2	16%	100%	16%	+3.7 cents: two direct lanes at 30% fill
4	Same-day ≥ 30%	$3.176	$2.412	1 (West)	44%	94%	44%	Whole trucks are lumpy: 30% costs the same as 44%
5	Same-day ≥ 50%	$3.235	$2.465	1	50%	94%	50%	+11 cents over no same-day
6	At most 1 hub	$3.176	$2.412	1	44%	94%	44%	One hub plus direct lanes: 5 cents more, 6% lose next-day
7	All 5 hubs forced open	$3.533	$2.762	5	0%	100%	0%	Three idle hubs cost 41 cents a package
8	Hubs only, same-day ≥ 15%	infeasible						Same-day cannot be bought with hubs
9	Peak week, +25% demand	$3.113	$2.341	2	21%	96%	21%	Hubs full; overflow goes direct
10	Peak + expand Hub West (×1.5, +$1,500 a week)	$3.132	$2.360	2	5%	100%	5%	Saves $834 of trucks, costs $1,500: does not pay; break-even $834

11	Dedicated fleet, cost per mile −20% (scenario 3 service)	$3.088	$2.324	2	16%	100%	16%	7 cents a package
12	Average hours ≤ 6.0	$3.176	$2.412	1	44%	94%	44%	A speed target is bought the same way as same-day

Reading the scenarios, in the order an interviewer would ask about them.

- **Why two hubs and nothing direct (1 and 2)?** At 4,000 packages a day every lane is thin, so Section 8.5's break-even is never reached and every zone rides a hub. Two hubs, not one, because the region is wide and the second hub saves more outbound miles than its $4,000 a week costs. Next-day costs nothing extra because every hub path is already under 9.5 hours; the rule is met without being paid for. A service rule that costs nothing is still worth writing down, because it tells you how much geography you can add before it starts to cost.
- **Why does 30% same-day cost the same as 44% (4)?** Because trucks come whole. To give Downtown same-day you must run a daily direct truck, and once it runs, every Downtown package rides it, so same-day jumps from 16% to 44% in one step. The cost–service curve is a staircase, not a ramp, and the planner should put the target on a step, not between steps where you pay for service you do not get.
- **Why do five hubs cost 41 cents (7)?** The three extra buildings add $11,600 a week of fixed cost (Metro $5,000, North $3,000, South $3,600), and $11,600 ÷ 28,000 packages is 41 cents. They save almost nothing in miles: linehaul falls from 28.8 to 28.5 cents a package, because the two chosen hubs were already well placed. This is the number to quote when someone proposes a building "for resilience": resilience is real, and here is its price per package.
- **Why is 8 infeasible?** Through a hub the fastest path is drive + 3-hour window + drive, which is over 4 hours at any distance. No mix of hubs can produce same-day. Infeasible is a useful answer: it says the target needs a different kind of path, not a better arrangement of the same kind.
- **Why does the hub expansion not pay (10)?** At peak the hubs fill and the overflow rides direct trucks that are mostly empty, which costs $834 a week more than if the hub could take it. Expanding the hub costs $1,500 a week. So the answer is to let peak overflow go direct, and the break-even for the expansion is written down: it pays if the peak overflow grows to about $1,500 of trucks a week, roughly double. A network designer's recommendations should always carry the number at which they flip.
- **Why is a cheaper mile worth 7 cents (11)?** Because linehaul is 36 cents of scenario 3's $2.40 dock-to-door, and 20% of 36 cents is 7. That is the whole value of a 20% cheaper mile on this region, and it is the number to compare with the fixed cost of owning the fleet that delivers it.

The cost–service frontier: what each level of same-day costs (next-day ≥ 90% throughout)

SAME-DAY TARGET	0%	10%	20%	30%	40%	50%	60%	70%	80%
Cost per package	$3.123	$3.160	$3.176	$3.176	$3.176	$3.235	$3.275	$3.343	$3.418
Added cost	—	3.7¢	5.3¢	5.3¢	5.3¢	11.2¢	15.2¢	22.0¢	29.5¢
Hubs open	2	2	1	1	1	1	1	1	1
Fill on direct lanes	—	30%	17%	17%	17%	19%	19%	17%	18%

How to read the frontier in the room. The curve is flat, then steep. The first 15% of same-day is cheap because two zones are thick enough that their direct trucks run a third full. From 20% to 40% nothing changes because of the staircase: the same trucks carry it. Past 40%, each extra ten points needs another thin lane with a daily truck at a fifth full, and the cost climbs 4 to 7 cents a step. The direct-lane fill row explains the shape: it falls from 30% to 17% as thinner and thinner zones are given a truck. The business decision is where on this curve the extra revenue from a same-day promise stops covering the extra trucks, and that is a number marketing and finance supply; the model's job is to make the curve honest. A planner who brings the curve to the meeting, instead of one recommended point, lets the business choose, and that is the difference between advising and deciding.

IF SOMEONE ASKS "HOW WOULD YOU PLAN THE TRANSPORTATION?" — 232 WORDS

I start with what is in the trailer: forty packages to a gaylord, fifty-two gaylords to a trailer, two thousand packages, and it cubes out at a seventh of the weight limit, so I plan by positions, not pounds. I plan the fill as a choice about days: eighty percent overflows one day in a hundred and sixty, ninety one day in seven, and which is right depends on whether a later truck still makes the promise. Then what is in the mile: about two dollars forty to run the truck and eighty cents for the empty return, so three twenty a loaded mile, and I quote it with the empty share. Then each lane: packages a day, trucks a day, fill; a direct truck pays when the daily volume beats truck cost divided by what the hub path costs a package, about six hundred seventy a day on a hundred-and-forty-five-mile lane. Fleet is trips times cycle hours over twenty-four, plus ten percent, with trailers at double the tractors. Service is a constraint, bought with earlier departures, faster unloads, extra departures or direct trucks, each with a price. The scorecard is cost per mile, cost per package-mile, fill, miles, departures, the roll, on-time, dwell, empty miles, tender acceptance. And the workbook puts all of it in one model: minimize cost per package subject to same-day and next-day targets, and draws the curve.

8.10 A camera on the gaylord: machine learning that sees the air

What this is. Computer vision is a model that looks at a photograph and returns a measurement. This section is one transportation use case, and only one: a camera over an open gaylord that measures how much air is left in the box. The barcode scan already counts packages. It cannot see the air. Every number below is an example built from this packet's own gaylord (Section 8.2) and this packet's own $640 truck. It is not a system that has been operated.

Start with two boxes that the scan says are identical. Both have 30 packages. A person who only reads the scan would treat them the same. The photograph does not.

Two gaylords, 30 packages each

GAYLORD	PACKAGES THE SCAN COUNTED	FREIGHT HEIGHT IN THE PHOTO	WHAT THE PHOTO SAYS
G1	30	21.6 inches	The pile already reaches the plan line. Do not add more.
G2	30	14.4 inches	Same count, shorter pile. The packages are smaller, and the box was closed with room left.

That gap is the whole use case. A count of 30 cannot tell G1 from G2. A height can.

Where the 21.6-inch line comes from (the same gaylord as Section 8.2)

STEP	QUESTION	ARITHMETIC	ANSWER
1	How tall is the gaylord?	48 inches, from Section 8.2	**48 inches**
2	What share of the box does the plan use?	45 percent. The lid goes on when the truck is due, not when the box is physically full. Section 8.2.	**45%**
3	Plan height, if the top of the pile is level	0.45 × 48	**21.6 inches**
4	Check against the package count	45% of 53.3 cubic feet is about 24 cubic feet. At 0.6 cubic feet a package, 24 ÷ 0.6	**40 packages, the same 40 as Section 8.2**

Step 3 treats the top of the pile as level. A peaked pile makes one height look fuller than the box is. The worked examples below use a level top, and a peaked top is one of the cases the camera is told to hand back to a person.

G2, one step at a time: how much room is really there

STEP	QUESTION	ARITHMETIC	ANSWER
1	How many packages per inch are already in G2?	30 packages ÷ 14.4 inches	**2.083 packages per inch**
2	How many packages of that same size reach the plan line?	2.083 × 21.6 inches	**45 packages**
3	Room the photo found	45 − 30	**15 packages**
4	How many of those 15 are actually waiting for this destination?	The camera does not know this. The warehouse system does. On this example, 8 are waiting and 7 are not.	**8 packages you can add. The other 7 are real air with nothing to put in them.**

G1, for comparison: 30 packages ÷ 21.6 inches = 1.389 packages per inch, and 1.389 × 21.6 = 30. Room found = 30 − 30 = 0. Adding a package to G1 would push it past the line the building planned for.

Read the two steps as two different questions. The camera answers "how high is the pile?" The warehouse system answers "is there more freight for this destination still on the dock?" The saving exists only where both answers are yes. A photo of an early lid, with no freight waiting, saves nothing.

One day at this warehouse: 100,000 packages

STEP	QUESTION	ARITHMETIC	ANSWER
1	Gaylords in the day, at the plan of 40 packages each	100,000 ÷ 40	**2,500 gaylords**
2	Gaylords the camera flags: pile under the plan line, and more freight waiting	400 of the 2,500. 400 ÷ 2,500	**16%**
3	Packages you can actually add, 8 per flagged gaylord	400 × 8	**3,200 packages**
4	Trailer-trips those packages no longer need	3,200 ÷ 1,600. These are spread across the day's lanes, not two trucks on one lane.	**2 trailer-trips**
5	Dollars in a day	2 × $640	**$1,280**
6	Per package	$1,280 ÷ 100,000	**1.3 cents**
7	A year, on the same 360-day year as Chapter 4.2	$1,280 × 360	**$460,800, about $461,000**

The $461,000 is a model of this example, and only if the dock holds the lid and puts the 8 packages in. The camera does not save the money. The loader does, because the camera showed the air.

How the model is taught, and when it is allowed to speak

Three steps, and only the second one is the learning.

1. A fixed camera takes one photo of the open gaylord, from the same angle every time.

2. The model marks the rim of the box and the top of the freight. It learned that marking from example photos on which a person had already drawn both lines. An example teaching set is 2,000 such photos. The number 2,000 is an example of the size of the set, not a requirement.
3. The height in inches is then ordinary measurement: the distance between the two lines the model marked, scaled to the 48-inch box.

Copying the person's drawing is not the test that matters. A tape measure is. The rule used here: the camera's height has to land within 2 inches of the tape. Two inches on a 48-inch box is 2 ÷ 48 = 4.2 percent of the height.

The tape test, before anyone changes a lid

TEST	ARITHMETIC	RESULT
First round, 50 gaylords	46 within 2 inches of the tape. 46 ÷ 50	**92%. Below the bar. The camera may not advise the loader.**
The bar	95 of 100 within 2 inches	**95%**
After more labeled photos, 100 new gaylords	96 within 2 inches. 96 ÷ 100	**96%. The camera may light the gaylord. A person still closes the lid.**

The light has a definition. The flag line is 80 percent of the plan height: 0.80 × 21.6 = 17.28 inches, called 17.3. Amber means the pile is under 17.3 inches and the warehouse system shows more freight for that destination. Green means either the pile has reached 17.3 inches, or nothing more is waiting. The camera supplies the height. The color needs both.

On 100 gaylords the camera had never seen, checked against the tape

Truth is the tape: a gaylord is "early" if its freight is under 17.3 inches. The camera says "early" or "ok."

	TAPE SAYS EARLY	TAPE SAYS OK	ROW TOTAL
Camera says early	18	4	22
Camera says ok	2	76	78
Column total	20	80	100

QUESTION	ARITHMETIC	ANSWER
Of the boxes the camera flagged, how many were truly early?	18 ÷ 22	**81.8%. This is the precision.**
Of the boxes that were truly early, how many did the camera catch?	18 ÷ 20	**90%. This is the recall.**
Of the good boxes, how many did the camera flag by mistake?	4 ÷ 80	**5%**

The 4 false alarms are the expensive kind of mistake at a dock: a loader waits for freight that will not fit, and the truck's clock slips. That is why the first release lights the gaylord and a person still decides. The 2 misses are boxes with room that left anyway. 90 percent caught is the benefit; 5 percent false is the price. Both numbers go on the page, not only the flattering one.

What the camera is refused. A dark dock, a black bag against a black box, rain on the lens, and a peaked pile. Those photos are scored as "no measurement," and a person does the gaylord. The share of photos with no measurement is published next to the 96 percent, because a camera that skips the hard boxes and only reports the easy ones will show a fine accuracy and miss the air.

What this is not. A camera at the trailer door, counting gaylords, is a different use case and a weaker one: a barcode on each gaylord already counts boxes, and one photo from the door cannot see a hole hidden behind the rear wall. The scan's blind spot is the air inside the box. That is the photograph worth taking.

The scorecard row it creates. Share of gaylords closed under 17.3 inches while more freight for that destination was waiting. It sits with trailer fullness (Chapter 4.2) as an input. The outputs it should move are trailer-trips per day and the $0.40 a package. If the input falls and the trips do not, the dock is seeing the light and not holding the lid.

IF SOMEONE ASKS "WHERE WOULD YOU USE MACHINE LEARNING IN TRANSPORTATION?" — 151 WORDS

One camera, one job: the air inside an open gaylord, which the package count cannot see. Two boxes with thirty packages can be different. One already reaches the plan line of 21.6 inches, so it is full. The other is at 14.4 inches, and the same packages would reach that line at forty-five, so there is room for fifteen more. Only eight of the fifteen are actually waiting, so only eight count. Four hundred gaylords like that, on a day of a hundred thousand packages, is 3,200 packages, two trucks, $1,280, about 1.3 cents a package. I would not let the camera close the lid. First it has to match a tape within two inches on ninety-five boxes out of a hundred. On the test it caught eighteen of twenty early boxes and raised four false alarms out of eighty good ones, and both of those numbers stay on the page.

Chapter 9 — Plain words and interview words; what is real and what is made up

The plain words used here, and the words you will hear in the interview

THIS DOCUMENT SAID	THE INTERVIEW WILL SAY	NOTE
The warehouse	Fulfillment center	Where items are stored, picked, and packed.
The sorting hub	Sort center; sortation center	Amazon has described two kinds: hubs that fill long-distance trucks, and last-mile sort centers that feed delivery stations. Ours is the second kind.
The depot	Delivery station	Where packages are sorted to van routes.
Labeling (weigh, check, label, record)	SLAM — scan, label, apply, manifest	Amazon's public term for the labeling step.
The truck leaves at 9 PM; the truck's leave time	Critical pull time	The time a package must be on a truck and the truck must depart. The most-used term in this job. Defined in a 2016 academic thesis on Amazon; no official public standard.
The truck ride between buildings	Linehaul; middle mile	Warehouse to hub, hub to depot.
Van drivers from a contracted van company	Delivery Service Partners	Independent companies running vans for Amazon since 2018.
Afternoon gig drivers in their own cars	Amazon Flex	Blocks of three to six hours, pickup at a delivery station.
Outside carrier	Carrier; postal injection; third-party carrier	Postal service, UPS, and similar.
Arrived on the promised day	Delivery estimate accuracy; on-time delivery	Share of packages with a delivery attempt on or before the promised date.
Transportation cost per package; dock to door	Transportation cost per package; cost per package; cost to serve (broader)	Cost to serve usually includes the warehouse too. Say which you mean.
Sent to the wrong building	Missort	
Left behind at the depot	Rolled volume; rollover	Packages that missed the wave they arrived for.
Trailer fullness	Trailer utilization; fill rate; cube utilization	Fullness can be counted in positions, cubic feet, or weight; say which.
Packages per van	Route density; packages per route; stops per route	
The pile; the flow in and out; the deadline	Stock and flow; work in process; the cutoff	Pile tomorrow = pile today + in − out.
Time left when the pile hits zero	Slack; buffer; pack-complete slack to the critical pull time	
"9 trips in 10 are within 4 hours 27 minutes"	The 90th percentile; p90	Plan deadlines on this, not on the average.
Which road: through the hub, direct, carrier	Network path; ship method; injection strategy	Direct trucks from a fulfillment center to a delivery station are called direct injection.
Mix versus rate	Volume, rate, and mix variance; the bridge; the waterfall	Chapter 6.2.
The weekly scorecard and meeting	Weekly business review	
Gaylord; the trailer is full by space / by weight	Gaylord (also "GL" or bulk container); cube out / weigh out; cube utilization	Chapter 8.2. Always say which limit binds.
A camera that measures the air in an open gaylord	Computer vision; a fill model	Section 8.10. A design example, not a system in operation. The scan counts packages; the camera measures height.
Packages stacked loose, no containers	Fluid load; floor load	More packages, slower unload.
Cost per loaded mile; the empty return	Linehaul rate; all-in rate per mile; deadhead (empty miles); round-trip pricing	Chapter 8.4. Quote the rate with the empty share.
Our own trucks and drivers, kept busy	Dedicated fleet; private fleet	Cheaper per mile only when utilization is steady.
A carrier paid per load at a yearly bid rate; whether they take the load	Contract (for-hire) truckload carrier; lane bid / RFP; tender acceptance; spot market	Chapter 8.4 and the scorecard in 8.8.
One origin, one destination, a schedule	Lane; departures per day; frequency	Chapter 8.5.
Driver leaves the trailer and takes an empty one	Drop and hook (versus live unload)	Shortens the tractor cycle; needs spare trailers.
How long a driver may drive and work in a day	Hours of service: 11 hours driving, 14 on duty, 10 off	Chapter 8.6; United States federal rule.
Two drivers sharing a tractor; a fast premium service	Team drivers; expedited	Roughly double cost per mile.
Cost per package-mile	Cost per package-mile; cost per cube-mile; cost per hundredweight (CWT) in general freight	Chapter 8.8's best single efficiency number.
The cost–service curve	Efficient frontier; Pareto frontier; cost-to-serve curve	Chapter 8.9.
How full to plan the truck; the share of days it overflows	Planned fill / load factor; service-level quantile; newsvendor critical ratio; roll-over (rolled volume)	Chapter 8.3.
The range that holds 95 (or 99) days in 100	95% / 99% confidence interval; two-sided band (2.5% or 0.5% of days above the top) versus one-sided service level (5% or 1% of days above the line)	Chapter 8.3. Average ± 1.96 swings, and ± 2.58 swings.
Cases per layer and layers per pallet	Ti-hi (tier × height); pallet pattern	Chapter 8.2, inbound table.

What runs out first when loading a trailer	Binding constraint; cube-limited vs weight-limited; density break-even (about 15 pounds per cubic foot)	Chapter 8.2.
Transshipment problem; which buildings to open	Transshipment / network flow problem; facility location; mixed-integer linear program (MILP); Solver (Simplex LP with integer constraints)	The workbook's two models.

What is real and what is made up

Made up, all of it, for the arithmetic: every rate, time, price, count, and percentage in Chapters 1 to 7. The $22 an hour, the 160 items an hour, the 2,000-package trailer, the $3.20 a mile, the 12 doors, the $264 van, the 98.1%, the twelve cents. They are chosen to be the right size and to be easy to check, and they are stated in one place (Chapter 1) so that changing one changes everything consistently.

Real, and public: the shape of the journey and the names of the steps.

PUBLIC FACT	SOURCE
The path fulfillment center → sort center → delivery station; the labeling step called SLAM; lanes as scheduled trucks filled so that neither the truck leaves empty nor the dock is left holding packages. Amazon Science, October 2022.	https://www.amazon.science/blog/maximizing-the-efficiency-of-amazons-own-delivery-networks
Middle mile from fulfillment centers into sort centers, sorted by destination, then to delivery stations or carrier partners such as the local post office. Amazon, transportation network article.	https://www.aboutamazon.com/news/operations/powering-amazons-transportation-network
Critical pull time: the time an item must be picked, palletized, and on a truck. 2016 academic thesis on Amazon transshipment.	http://hdl.handle.net/1721.1/106075
Delivery estimate accuracy: share of items with a delivery attempt on or before the promised date; missort named as a cause; no rate published. 2024 academic thesis.	https://hdl.handle.net/1721.1/155603
Eight largely self-sufficient United States regions, replacing one national network; in-region fulfillment from 62% to 76%; direct fulfillment-center-to-delivery-station trucks became easier to fill. Amazon, 2023.	https://www.aboutamazon.com/news/company-news/amazon-ceo-andy-jassy-2022-letter-to-shareholders and https://www.amazon.science/news-and-features/how-amazon-reworked-its-fulfillment-network-to-meet-customer-demand
Middle-mile buildings specialized into hubs (long-haul fill) and last-mile sort centers (feeding delivery stations). Journal on Applied Analytics, 2025.	https://pubsonline.informs.org/doi/full/10.1287/inte.2025.0295
Delivery Service Partner program, launched 2018; Amazon Flex blocks of three to six hours with pickup at a delivery station.	https://www.aboutamazon.com/impact/empowerment/delivery-partners and https://flex.amazon.com/support/faq
Delivery dates shown to customers are the shipping date plus transit time; an "order within" countdown is shown when it applies.	https://www.amazon.com/gp/help/customer/display.html?nodeId=GV8H5D3MMAR7JBLF

Not public, and therefore not claimed here: any Amazon cost per package, trailer fullness, sorter speed, van cost, truck rate, or on-time percentage; any official standard for critical pull time; the names of any internal systems. When any of these comes up in the room, the honest sentence is: "In a reference network I built with stated assumptions, the number is X; I would not guess Amazon's."

How this companion relates to the detailed packet

The companion workbook *outbound-network-design-model.xlsx* holds the baseline transshipment model and the outbound model of Chapter 8.9, both set up for Excel's Solver, with the twelve scenarios and the frontier. The detailed packet in the same folder covers the same network with more: the eight job responsibilities worked one by one (new network designs, whether a sorting building earns its rent, the business case for a new building, how to run a pilot, how to write the recommendation), the candidate's own record mapped to each, and a formula sheet. Read this companion first. When the detailed packet uses a term, this chapter's first table translates it.